

```
In [4]: '''# -----
# small (t=5, seller=3, buyer=3, transporter=3)
sellers = {
    1: {"supply": 100, "price": [40, 35, 70, 0, 0]},
    2: {"supply": 120, "price": [0, 35, 38, 45, 50]},
    3: {"supply": 90, "price": [38, 0, 0, 42, 40]}
}

buyers = {
    1: {"demand": 80, "price": [70, 72, 68, 65, 60]},
    2: {"demand": 100, "price": [0, 65, 70, 0, 68]},
    3: {"demand": 60, "price": [68, 0, 0, 66, 70]}
}

transporters = {
    1: {"cap": {1:100,2:100,3:80,4:120,5:100}, "cost":5},
    2: {"cap": {1:50,2:50,3:60,4:50,5:70}, "cost":4},
    3: {"cap": {1:80,2:60,3:90,4:100,5:110}, "cost":4}
}

T = [1,2,3,4,5]
W = 3'''
```

```
Out[4]: '# -----\n# small (t=5, seller=3, buyer=3, transporter=3)\nsellers = {\n    1: {"supply": 100, "price": [40, 35, 70, 0, 0]},\n    2: {"supply": 120, "price": [0, 35, 38, 45, 50]},\n    3: {"supply": 90, "price": [38, 0, 0, 42, 40]}\n}\nbuyers = {\n    1: {"demand": 80, "price": [70, 72, 68, 65, 60]},\n    2: {"demand": 100, "price": [0, 65, 70, 0, 68]},\n    3: {"demand": 60, "price": [68, 0, 0, 66, 70]}\n}\ntransporters = {\n    1: {"cap": {1:100,2:100,3:80,4:120,5:100}, "cost":5},\n    2: {"cap": {1:50,2:50,3:60,4:50,5:70}, "cost":4},\n    3: {"cap": {1:80,2:60,3:90,4:100,5:110}, "cost":4}\n}\nT = [1,2,3,4,5]\nW = 3'
```

```
In [5]: '''# -----
# middle (T=10, seller=40, buyer=40, transporter=20)
sellers = {
    1: {"supply": 124, "price": [45, 0, 52, 60, 58, 0, 64, 59, 61, 55]},
    2: {"supply": 97, "price": [0, 48, 51, 54, 0, 50, 49, 0, 53, 47]},
    3: {"supply": 142, "price": [40, 45, 0, 42, 43, 44, 0, 46, 48, 49]},
    4: {"supply": 120, "price": [0, 38, 41, 43, 0, 46, 45, 47, 50, 52]},
    5: {"supply": 133, "price": [35, 0, 39, 40, 0, 42, 43, 44, 46, 0]},
    6: {"supply": 106, "price": [0, 34, 36, 0, 38, 40, 42, 44, 45, 0]},
    7: {"supply": 128, "price": [50, 52, 0, 54, 56, 57, 0, 59, 0, 60]},
    8: {"supply": 114, "price": [45, 47, 0, 48, 49, 0, 51, 52, 54, 55]},
    9: {"supply": 136, "price": [0, 43, 45, 0, 48, 50, 0, 52, 53, 55]},
    10: {"supply": 89, "price": [35, 37, 0, 38, 39, 0, 41, 0, 43, 44]},
    11: {"supply": 141, "price": [55, 57, 58, 0, 60, 61, 0, 63, 64, 0]},
    12: {"supply": 115, "price": [42, 0, 44, 45, 46, 0, 47, 48, 49, 0]},
    13: {"supply": 124, "price": [39, 40, 0, 41, 0, 42, 43, 44, 45, 46]},
    14: {"supply": 132, "price": [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]},
    15: {"supply": 107, "price": [0, 36, 38, 39, 0, 40, 41, 42, 43, 44]},
    16: {"supply": 120, "price": [49, 0, 51, 0, 52, 54, 0, 56, 57, 0]},
    17: {"supply": 103, "price": [40, 42, 0, 43, 44, 45, 46, 0, 47, 0]},
    18: {"supply": 129, "price": [0, 50, 52, 53, 0, 55, 56, 57, 58, 0]},
    19: {"supply": 118, "price": [37, 38, 0, 39, 40, 41, 42, 0, 43, 44]},
    20: {"supply": 125, "price": [0, 45, 46, 47, 48, 0, 49, 50, 51, 0]},
    21: {"supply": 132, "price": [55, 0, 57, 58, 59, 0, 60, 61, 0, 63]},
    22: {"supply": 139, "price": [40, 41, 0, 42, 43, 0, 44, 45, 0, 46]},
}
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23: {"supply": 127, "price": [0, 48, 49, 50, 51, 52, 0, 53, 54, 0]},
24: {"supply": 133, "price": [0, 37, 38, 39, 40, 41, 42, 0, 43, 0]},
25: {"supply": 144, "price": [42, 0, 43, 44, 0, 46, 47, 48, 49, 0]},
26: {"supply": 95, "price": [0, 50, 52, 53, 54, 0, 55, 56, 57, 0]},
27: {"supply": 130, "price": [39, 0, 40, 41, 42, 43, 0, 44, 45, 0]},
28: {"supply": 138, "price": [0, 47, 48, 0, 50, 51, 52, 0, 53, 54]},
29: {"supply": 101, "price": [36, 37, 38, 0, 39, 0, 40, 41, 0, 42]},
30: {"supply": 119, "price": [0, 44, 45, 46, 47, 48, 0, 49, 50, 0]},
31: {"supply": 140, "price": [51, 0, 52, 53, 54, 0, 55, 56, 57, 0]},
32: {"supply": 111, "price": [0, 42, 43, 0, 44, 45, 46, 47, 0, 48]},
33: {"supply": 108, "price": [39, 0, 40, 41, 42, 43, 0, 44, 45, 46]},
34: {"supply": 136, "price": [0, 49, 50, 51, 52, 0, 53, 54, 0, 55]},
35: {"supply": 122, "price": [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]},
36: {"supply": 118, "price": [0, 35, 36, 37, 0, 38, 39, 40, 41, 0]},
37: {"supply": 131, "price": [40, 41, 42, 0, 43, 44, 0, 45, 46, 0]},
38: {"supply": 100, "price": [0, 45, 47, 0, 48, 49, 50, 0, 51, 52]},
39: {"supply": 129, "price": [37, 0, 38, 39, 40, 0, 41, 42, 43, 0]},
40: {"supply": 105, "price": [0, 53, 54, 0, 55, 56, 57, 0, 58, 59]}
}

buyers = {
  1: {"demand": 90, "price": [75, 76, 0, 78, 79, 0, 81, 82, 83, 0]},
  2: {"demand": 100, "price": [0, 68, 70, 72, 0, 74, 75, 0, 77, 78]},
  3: {"demand": 85, "price": [0, 66, 68, 0, 70, 71, 72, 73, 74, 0]},
  4: {"demand": 95, "price": [80, 0, 81, 82, 83, 0, 85, 86, 0, 87]},
  5: {"demand": 100, "price": [0, 70, 72, 73, 74, 75, 0, 77, 78, 0]},
  6: {"demand": 110, "price": [68, 69, 0, 71, 72, 73, 0, 75, 76, 0]},
  7: {"demand": 78, "price": [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]},
  8: {"demand": 105, "price": [82, 83, 0, 84, 0, 85, 86, 0, 87, 88]},
  9: {"demand": 115, "price": [0, 67, 69, 70, 71, 0, 72, 73, 0, 74]},
  10: {"demand": 98, "price": [75, 0, 76, 77, 0, 78, 79, 80, 0, 81]},
  11: {"demand": 120, "price": [0, 70, 72, 0, 74, 75, 76, 77, 78, 0]},
  12: {"demand": 86, "price": [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]},
  13: {"demand": 92, "price": [0, 65, 67, 0, 69, 70, 71, 72, 73, 0]},
  14: {"demand": 88, "price": [76, 77, 78, 0, 79, 80, 81, 0, 82, 83]},
  15: {"demand": 113, "price": [0, 72, 73, 74, 0, 75, 76, 77, 0, 78]},
  16: {"demand": 109, "price": [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]},
  17: {"demand": 91, "price": [0, 69, 70, 0, 72, 73, 74, 75, 0, 76]},
  18: {"demand": 104, "price": [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]},
  19: {"demand": 97, "price": [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]},
  20: {"demand": 118, "price": [0, 66, 68, 69, 0, 70, 71, 72, 73, 0]},
  21: {"demand": 99, "price": [83, 0, 84, 85, 0, 86, 87, 88, 0, 89]},
  22: {"demand": 84, "price": [0, 70, 72, 73, 0, 74, 75, 76, 0, 77]},
  23: {"demand": 110, "price": [80, 0, 81, 82, 83, 0, 84, 85, 86, 0]},
  24: {"demand": 93, "price": [0, 69, 70, 0, 71, 72, 73, 74, 75, 0]},
  25: {"demand": 102, "price": [77, 78, 0, 79, 0, 80, 81, 82, 0, 83]},
  26: {"demand": 108, "price": [0, 73, 74, 75, 0, 76, 77, 0, 78, 79]},
  27: {"demand": 96, "price": [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]},
  28: {"demand": 87, "price": [0, 65, 67, 68, 0, 70, 71, 72, 0, 73]},
  29: {"demand": 116, "price": [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]},
  30: {"demand": 90, "price": [0, 68, 69, 0, 70, 71, 72, 73, 0, 74]},
  31: {"demand": 114, "price": [84, 0, 85, 86, 0, 87, 88, 89, 0, 90]},
  32: {"demand": 92, "price": [0, 70, 72, 73, 74, 0, 75, 76, 0, 77]},
  33: {"demand": 106, "price": [78, 79, 0, 80, 81, 0, 82, 83, 0, 84]},
  34: {"demand": 120, "price": [0, 66, 67, 68, 69, 0, 70, 71, 0, 72]},
  35: {"demand": 100, "price": [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]},
  36: {"demand": 115, "price": [0, 69, 71, 0, 72, 73, 74, 75, 0, 76]},
  37: {"demand": 119, "price": [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]},
  38: {"demand": 98, "price": [0, 70, 71, 72, 0, 73, 74, 75, 0, 76]},
  39: {"demand": 112, "price": [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]},

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    40: {"demand": 94,  "price": [0, 68, 69, 70, 0, 71, 72, 73, 74, 0]}
}

transporters = {
    1: {"cap": {1:120,2:100,3:90,4:110,5:100,6:130,7:120,8:140,9:100,10:120}, "c
    2: {"cap": {1:80,2:70,3:60,4:90,5:80,6:100,7:85,8:95,9:70,10:80}, "cost":5},
    3: {"cap": {1:100,2:120,3:110,4:130,5:120,6:140,7:135,8:125,9:120,10:115}, "
    4: {"cap": {1:90,2:100,3:80,4:110,5:120,6:130,7:120,8:140,9:110,10:120}, "co
    5: {"cap": {1:70,2:60,3:80,4:75,5:85,6:90,7:95,8:80,9:85,10:90}, "cost":3},
    6: {"cap": {1:100,2:120,3:130,4:110,5:125,6:140,7:135,8:120,9:125,10:130}, "
    7: {"cap": {1:85,2:90,3:95,4:100,5:105,6:110,7:115,8:120,9:125,10:130}, "cos
    8: {"cap": {1:60,2:65,3:70,4:75,5:80,6:85,7:90,8:95,9:100,10:105}, "cost":3}
    9: {"cap": {1:110,2:120,3:130,4:125,5:115,6:140,7:135,8:145,9:130,10:120}, "
    10: {"cap": {1:100,2:105,3:110,4:115,5:120,6:125,7:130,8:135,9:140,10:145},
    11: {"cap": {1:75,2:80,3:85,4:90,5:95,6:100,7:105,8:110,9:115,10:120}, "cost
    12: {"cap": {1:90,2:100,3:110,4:120,5:130,6:140,7:120,8:110,9:100,10:90}, "c
    13: {"cap": {1:60,2:70,3:80,4:90,5:100,6:110,7:120,8:130,9:140,10:150}, "cos
    14: {"cap": {1:130,2:125,3:120,4:115,5:110,6:105,7:100,8:95,9:90,10:85}, "co
    15: {"cap": {1:100,2:95,3:90,4:85,5:80,6:75,7:70,8:65,9:60,10:55}, "cost":3}
    16: {"cap": {1:90,2:100,3:110,4:120,5:130,6:140,7:135,8:125,9:115,10:105}, "
    17: {"cap": {1:80,2:85,3:90,4:95,5:100,6:105,7:110,8:115,9:120,10:125}, "cos
    18: {"cap": {1:100,2:110,3:120,4:130,5:140,6:150,7:135,8:125,9:115,10:105},
    19: {"cap": {1:95,2:90,3:85,4:100,5:110,6:115,7:120,8:125,9:130,10:135}, "co
    20: {"cap": {1:70,2:80,3:90,4:100,5:110,6:120,7:130,8:140,9:150,10:160}, "co
}

T = [1,2,3,4,5,6,7,8,9,10]'''

```

```

Out[5]: '# -----\n# middle (T=10, seller=40, buyer=40, transporter=20)\nnsellers = {\n    1: {"supply": 124, "price": [45, 0, 52, 60, 58, 0, 64, 59, 61, 55]},\n    2: {"supply": 97, "price": [0, 48, 51, 54, 0, 50, 49, 0, 53, 47]},\n    3: {"supply": 142, "price": [40, 45, 0, 42, 43, 44, 0, 46, 48, 49]},\n    4: {"supply": 120, "price": [0, 38, 41, 43, 0, 46, 45, 47, 50, 52]},\n    5: {"supply": 133, "price": [35, 0, 39, 40, 0, 42, 43, 44, 46, 0]},\n    6: {"supply": 106, "price": [0, 34, 36, 0, 38, 40, 42, 44, 45, 0]},\n    7: {"supply": 128, "price": [50, 52, 0, 54, 56, 57, 0, 59, 0, 60]},\n    8: {"supply": 114, "price": [45, 47, 0, 48, 49, 0, 51, 52, 54, 55]},\n    9: {"supply": 136, "price": [0, 43, 45, 0, 48, 50, 0, 52, 53, 55]},\n    10: {"supply": 89, "price": [35, 37, 0, 38, 39, 0, 41, 0, 43, 44]},\n    11: {"supply": 141, "price": [55, 57, 58, 0, 60, 61, 0, 63, 64, 0]},\n    12: {"supply": 115, "price": [42, 0, 44, 45, 46, 0, 47, 48, 49, 0]},\n    13: {"supply": 124, "price": [39, 40, 0, 41, 0, 42, 43, 44, 45, 46]},\n    14: {"supply": 132, "price": [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]},\n    15: {"supply": 107, "price": [0, 36, 38, 39, 0, 40, 41, 42, 43, 44]},\n    16: {"supply": 120, "price": [49, 0, 51, 0, 52, 54, 0, 56, 57, 0]},\n    17: {"supply": 103, "price": [40, 42, 0, 43, 44, 45, 46, 0, 47, 0]},\n    18: {"supply": 129, "price": [0, 50, 52, 53, 0, 55, 56, 57, 58, 0]},\n    19: {"supply": 118, "price": [37, 38, 0, 39, 40, 41, 42, 0, 43, 44]},\n    20: {"supply": 125, "price": [0, 45, 46, 47, 48, 0, 49, 50, 51, 0]},\n    21: {"supply": 132, "price": [55, 0, 57, 58, 59, 0, 60, 61, 0, 63]},\n    22: {"supply": 139, "price": [40, 41, 0, 42, 43, 0, 44, 45, 0, 46]},\n    23: {"supply": 127, "price": [0, 48, 49, 50, 51, 52, 0, 53, 54, 0]},\n    24: {"supply": 133, "price": [0, 37, 38, 39, 40, 41, 42, 0, 43, 0]},\n    25: {"supply": 144, "price": [42, 0, 43, 44, 0, 46, 47, 48, 49, 0]},\n    26: {"supply": 95, "price": [0, 50, 52, 53, 54, 0, 55, 56, 57, 0]},\n    27: {"supply": 130, "price": [39, 0, 40, 41, 42, 43, 0, 44, 45, 0]},\n    28: {"supply": 138, "price": [0, 47, 48, 0, 50, 51, 52, 0, 53, 54]},\n    29: {"supply": 101, "price": [36, 37, 38, 0, 39, 0, 40, 41, 0, 42]},\n    30: {"supply": 119, "price": [0, 44, 45, 46, 47, 48, 0, 49, 50, 0]},\n    31: {"supply": 140, "price": [51, 0, 52, 53, 54, 0, 55, 56, 57, 0]},\n    32: {"supply": 111, "price": [0, 42, 43, 0, 44, 45, 46, 47, 0, 48]},\n    33: {"supply": 108, "price": [39, 0, 40, 41, 42, 43, 0, 44, 45, 46]},\n    34: {"supply": 136, "price": [0, 49, 50, 51, 52, 0, 53, 54, 0, 55]},\n    35: {"supply": 122, "price": [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]},\n    36: {"supply": 118, "price": [0, 35, 36, 37, 0, 38, 39, 40, 41, 0]},\n    37: {"supply": 131, "price": [40, 41, 42, 0, 43, 44, 0, 45, 46, 0]},\n    38: {"supply": 100, "price": [0, 45, 47, 0, 48, 49, 50, 0, 51, 52]},\n    39: {"supply": 129, "price": [37, 0, 38, 39, 40, 0, 41, 42, 43, 0]},\n    40: {"supply": 105, "price": [0, 53, 54, 0, 55, 56, 57, 0, 58, 59]}\n}\n\nnbuyers = {\n    1: {"demand": 90, "price": [75, 76, 0, 78, 79, 0, 81, 82, 83, 0]},\n    2: {"demand": 100, "price": [0, 68, 70, 72, 0, 74, 75, 0, 77, 78]},\n    3: {"demand": 85, "price": [0, 66, 68, 0, 70, 71, 72, 73, 74, 0]},\n    4: {"demand": 95, "price": [80, 0, 81, 82, 83, 0, 85, 86, 0, 87]},\n    5: {"demand": 100, "price": [0, 70, 72, 73, 74, 75, 0, 77, 78, 0]},\n    6: {"demand": 110, "price": [68, 69, 0, 71, 72, 73, 0, 75, 76, 0]},\n    7: {"demand": 78, "price": [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]},\n    8: {"demand": 105, "price": [82, 83, 0, 84, 0, 85, 86, 0, 87, 88]},\n    9: {"demand": 115, "price": [0, 67, 69, 70, 71, 0, 72, 73, 0, 74]},\n    10: {"demand": 98, "price": [75, 0, 76, 77, 0, 78, 79, 80, 0, 81]},\n    11: {"demand": 120, "price": [0, 70, 72, 0, 74, 75, 76, 77, 78, 0]},\n    12: {"demand": 86, "price": [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]},\n    13: {"demand": 92, "price": [0, 65, 67, 0, 69, 70, 71, 72, 73, 0]},\n    14: {"demand": 88, "price": [76, 77, 78, 0, 79, 80, 81, 0, 82, 83]},\n    15: {"demand": 113, "price": [0, 72, 73, 74, 0, 75, 76, 77, 0, 78]},\n    16: {"demand": 109, "price": [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]},\n    17: {"demand": 91, "price": [0, 69, 70, 0, 72, 73, 74, 75, 0, 76]},\n    18: {"demand": 104, "price": [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]},\n    19: {"demand": 97, "price": [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]},\n    20: {"demand": 118, "price": [0, 66, 68, 69, 0, 70, 71, 72, 73, 0]},\n    21: {"demand": 99, "price": [83, 0, 84, 85, 0, 86, 87, 88, 0, 89]},\n    22: {"demand": 84, "p

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rice": [0, 70, 72, 73, 0, 74, 75, 76, 0, 77]],\n    23: {"demand": 110, "price": [80, 0, 81, 82, 83, 0, 84, 85, 86, 0]],\n    24: {"demand": 93, "price": [0, 69, 70, 0, 71, 72, 73, 74, 75, 0]],\n    25: {"demand": 102, "price": [77, 78, 0, 79, 0, 80, 81, 82, 0, 83]],\n    26: {"demand": 108, "price": [0, 73, 74, 75, 0, 76, 77, 0, 78, 79]],\n    27: {"demand": 96, "price": [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]],\n    28: {"demand": 87, "price": [0, 65, 67, 68, 0, 70, 71, 72, 0, 73]],\n    29: {"demand": 116, "price": [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]],\n    30: {"demand": 90, "price": [0, 68, 69, 0, 70, 71, 72, 73, 0, 74]],\n    31: {"demand": 114, "price": [84, 0, 85, 86, 0, 87, 88, 89, 0, 90]],\n    32: {"demand": 92, "price": [0, 70, 72, 73, 74, 0, 75, 76, 0, 77]],\n    33: {"demand": 106, "price": [78, 79, 0, 80, 81, 0, 82, 83, 0, 84]],\n    34: {"demand": 120, "price": [0, 66, 67, 68, 69, 0, 70, 71, 0, 72]],\n    35: {"demand": 100, "price": [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]],\n    36: {"demand": 115, "price": [0, 69, 71, 0, 72, 73, 74, 75, 0, 76]],\n    37: {"demand": 119, "price": [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]],\n    38: {"demand": 98, "price": [0, 70, 71, 72, 0, 73, 74, 75, 0, 76]],\n    39: {"demand": 112, "price": [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]],\n    40: {"demand": 94, "price": [0, 68, 69, 70, 0, 71, 72, 73, 74, 0]]}\n\nntransporters = {\n    1: {"cap": {1:120,2:100,3:90,4:110,5:100,6:130,7:120,8:140,9:100,10:120}, "cost":4},\n    2: {"cap": {1:80,2:70,3:60,4:90,5:80,6:100,7:85,8:95,9:70,10:80}, "cost":5},\n    3: {"cap": {1:100,2:120,3:110,4:130,5:120,6:140,7:135,8:125,9:120,10:115}, "cost":6},\n    4: {"cap": {1:90,2:100,3:80,4:110,5:120,6:130,7:120,8:140,9:110,10:120}, "cost":4},\n    5: {"cap": {1:70,2:60,3:80,4:75,5:85,6:90,7:95,8:80,9:85,10:90}, "cost":3},\n    6: {"cap": {1:100,2:120,3:130,4:110,5:125,6:140,7:135,8:120,9:125,10:130}, "cost":6},\n    7: {"cap": {1:85,2:90,3:95,4:100,5:105,6:110,7:115,8:120,9:125,10:130}, "cost":5},\n    8: {"cap": {1:60,2:65,3:70,4:75,5:80,6:85,7:90,8:95,9:100,10:105}, "cost":3},\n    9: {"cap": {1:110,2:120,3:130,4:125,5:115,6:140,7:135,8:145,9:130,10:120}, "cost":6},\n    10: {"cap": {1:100,2:105,3:110,4:115,5:120,6:125,7:130,8:135,9:140,10:145}, "cost":5},\n    11: {"cap": {1:75,2:80,3:85,4:90,5:95,6:100,7:105,8:110,9:115,10:120}, "cost":4},\n    12: {"cap": {1:90,2:100,3:110,4:120,5:130,6:140,7:120,8:110,9:100,10:90}, "cost":5},\n    13: {"cap": {1:60,2:70,3:80,4:90,5:100,6:110,7:120,8:130,9:140,10:150}, "cost":4},\n    14: {"cap": {1:130,2:125,3:120,4:115,5:110,6:105,7:100,8:95,9:90,10:85}, "cost":6},\n    15: {"cap": {1:100,2:95,3:90,4:85,5:80,6:75,7:70,8:65,9:60,10:55}, "cost":3},\n    16: {"cap": {1:90,2:100,3:110,4:120,5:130,6:140,7:135,8:125,9:115,10:105}, "cost":5},\n    17: {"cap": {1:80,2:85,3:90,4:95,5:100,6:105,7:110,8:115,9:120,10:125}, "cost":4},\n    18: {"cap": {1:100,2:110,3:120,4:130,5:140,6:150,7:135,8:125,9:115,10:105}, "cost":6},\n    19: {"cap": {1:95,2:90,3:85,4:100,5:110,6:115,7:120,8:125,9:130,10:135}, "cost":4},\n    20: {"cap": {1:70,2:80,3:90,4:100,5:110,6:120,7:130,8:140,9:150,10:160}, "cost":5}}\n\nnT = [1,2,3,4,5,6,7,8,9,10]'

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In [6]: # -----
# large (T=10, seller=100, buyer=100, transporter=50)
sellers = {
    1: {"supply": 124, "price": [45, 0, 52, 40, 58, 0, 64, 59, 61, 55]},
    2: {"supply": 97, "price": [0, 48, 51, 54, 0, 50, 49, 0, 53, 47]},
    3: {"supply": 142, "price": [40, 45, 0, 42, 43, 44, 0, 46, 48, 49]},
    4: {"supply": 120, "price": [0, 38, 41, 43, 0, 46, 45, 47, 50, 52]},
    5: {"supply": 133, "price": [35, 0, 39, 40, 0, 42, 43, 44, 46, 0]},
    6: {"supply": 106, "price": [0, 34, 36, 0, 38, 40, 42, 44, 45, 0]},
    7: {"supply": 128, "price": [50, 52, 0, 54, 56, 57, 0, 59, 0, 60]},
    8: {"supply": 114, "price": [45, 47, 0, 48, 49, 0, 51, 52, 54, 55]},
    9: {"supply": 136, "price": [0, 43, 45, 0, 48, 50, 0, 52, 53, 55]},
    10: {"supply": 89, "price": [35, 37, 0, 38, 39, 0, 41, 0, 43, 44]},
    11: {"supply": 141, "price": [55, 57, 58, 0, 60, 61, 0, 63, 64, 0]},
    12: {"supply": 115, "price": [42, 0, 44, 45, 46, 0, 47, 48, 49, 0]},
    13: {"supply": 124, "price": [39, 40, 0, 41, 0, 42, 43, 44, 45, 46]},
    14: {"supply": 132, "price": [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]},
    15: {"supply": 107, "price": [0, 36, 38, 39, 0, 40, 41, 42, 43, 44]},

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16: {"supply": 120, "price": [49, 0, 51, 0, 52, 54, 0, 56, 57, 0]},
17: {"supply": 103, "price": [40, 42, 0, 43, 44, 45, 46, 0, 47, 0]},
18: {"supply": 129, "price": [0, 50, 52, 53, 0, 55, 56, 57, 58, 0]},
19: {"supply": 118, "price": [37, 38, 0, 39, 40, 41, 42, 0, 43, 44]},
20: {"supply": 125, "price": [0, 45, 46, 47, 48, 0, 49, 50, 51, 0]},
21: {"supply": 132, "price": [55, 0, 57, 50, 59, 0, 60, 61, 0, 63]},
22: {"supply": 139, "price": [40, 41, 0, 42, 43, 0, 44, 45, 0, 46]},
23: {"supply": 127, "price": [0, 48, 49, 50, 51, 52, 0, 53, 54, 0]},
24: {"supply": 133, "price": [0, 37, 38, 39, 40, 41, 42, 0, 43, 0]},
25: {"supply": 144, "price": [42, 0, 43, 44, 0, 46, 47, 48, 49, 0]},
26: {"supply": 95, "price": [0, 50, 52, 53, 54, 0, 55, 56, 57, 0]},
27: {"supply": 130, "price": [39, 0, 40, 41, 42, 43, 0, 44, 45, 0]},
28: {"supply": 138, "price": [0, 47, 48, 0, 50, 51, 52, 0, 53, 54]},
29: {"supply": 101, "price": [36, 37, 38, 0, 39, 0, 40, 41, 0, 42]},
30: {"supply": 119, "price": [0, 44, 45, 46, 47, 48, 0, 49, 50, 0]},
31: {"supply": 140, "price": [51, 0, 52, 53, 54, 0, 55, 56, 57, 0]},
32: {"supply": 111, "price": [0, 42, 43, 0, 44, 45, 46, 47, 0, 48]},
33: {"supply": 108, "price": [39, 0, 40, 41, 42, 43, 0, 44, 45, 46]},
34: {"supply": 136, "price": [0, 49, 50, 51, 52, 0, 53, 54, 0, 55]},
35: {"supply": 122, "price": [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]},
36: {"supply": 118, "price": [0, 35, 36, 37, 0, 38, 39, 40, 41, 0]},
37: {"supply": 131, "price": [40, 41, 42, 0, 43, 44, 0, 45, 46, 0]},
38: {"supply": 100, "price": [0, 45, 47, 0, 48, 49, 50, 0, 51, 52]},
39: {"supply": 129, "price": [37, 0, 38, 39, 40, 0, 41, 42, 43, 0]},
40: {"supply": 105, "price": [0, 53, 54, 0, 55, 56, 57, 0, 58, 59]},
41: {"supply": 117, "price": [44, 0, 45, 46, 47, 0, 48, 49, 50, 0]},
42: {"supply": 125, "price": [0, 38, 39, 40, 41, 0, 42, 43, 0, 44]},
43: {"supply": 130, "price": [50, 0, 52, 0, 53, 54, 0, 55, 56, 57]},
44: {"supply": 126, "price": [0, 42, 43, 44, 0, 45, 46, 0, 47, 48]},
45: {"supply": 121, "price": [37, 38, 0, 39, 0, 40, 41, 42, 43, 0]},
46: {"supply": 110, "price": [0, 47, 48, 49, 50, 0, 51, 52, 0, 53]},
47: {"supply": 119, "price": [45, 0, 46, 47, 48, 0, 49, 50, 51, 0]},
48: {"supply": 123, "price": [0, 39, 40, 41, 0, 42, 43, 44, 0, 45]},
49: {"supply": 129, "price": [41, 0, 43, 44, 0, 45, 46, 47, 0, 48]},
50: {"supply": 134, "price": [0, 49, 50, 51, 0, 52, 53, 54, 55, 0]},
51: {"supply": 100, "price": [40, 35, 38, 0, 42, 45, 50, 0, 55, 48]},
52: {"supply": 105, "price": [0, 36, 0, 41, 44, 40, 0, 47, 49, 42]},
53: {"supply": 110, "price": [38, 0, 42, 45, 0, 48, 50, 52, 0, 46]},
54: {"supply": 115, "price": [35, 37, 39, 0, 44, 0, 50, 53, 55, 0]},
55: {"supply": 120, "price": [0, 36, 38, 40, 42, 0, 48, 50, 52, 54]},
56: {"supply": 100, "price": [42, 0, 40, 43, 0, 47, 45, 0, 50, 48]},
57: {"supply": 105, "price": [0, 39, 0, 41, 44, 46, 0, 49, 51, 0]},
58: {"supply": 110, "price": [38, 36, 0, 0, 42, 45, 47, 0, 50, 53]},
59: {"supply": 115, "price": [35, 0, 39, 42, 0, 46, 48, 50, 0, 52]},
60: {"supply": 120, "price": [0, 36, 38, 41, 43, 0, 47, 49, 51, 0]},
61: {"supply": 100, "price": [40, 0, 37, 39, 42, 0, 45, 47, 0, 50]},
62: {"supply": 105, "price": [0, 36, 38, 0, 41, 44, 0, 46, 48, 50]},
63: {"supply": 110, "price": [39, 0, 42, 45, 0, 47, 50, 52, 0, 49]},
64: {"supply": 115, "price": [0, 37, 39, 41, 0, 44, 46, 48, 0, 51]},
65: {"supply": 120, "price": [38, 0, 0, 42, 45, 0, 47, 49, 51, 53]},
66: {"supply": 100, "price": [40, 35, 0, 43, 0, 46, 48, 50, 0, 52]},
67: {"supply": 105, "price": [0, 36, 38, 40, 0, 44, 46, 0, 49, 51]},
68: {"supply": 110, "price": [37, 0, 39, 41, 0, 45, 47, 49, 0, 50]},
69: {"supply": 115, "price": [0, 36, 38, 0, 42, 44, 46, 48, 0, 51]},
70: {"supply": 120, "price": [38, 0, 40, 42, 0, 45, 47, 0, 50, 52]},
71: {"supply": 100, "price": [39, 0, 37, 0, 42, 44, 46, 48, 0, 50]},
72: {"supply": 105, "price": [0, 36, 38, 40, 0, 43, 45, 0, 47, 49]},
73: {"supply": 110, "price": [38, 0, 41, 43, 0, 46, 48, 50, 0, 52]},
74: {"supply": 115, "price": [0, 37, 39, 0, 44, 46, 0, 48, 50, 52]},
75: {"supply": 120, "price": [36, 0, 38, 41, 0, 45, 47, 0, 49, 51]},

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76: {"supply":100, "price":[40,35,0,42,44,0,46,48,50,0]},
77: {"supply":105, "price":[0,36,38,0,41,43,45,0,47,49]},
78: {"supply":110, "price":[38,0,40,42,0,45,47,49,0,51]},
79: {"supply":115, "price":[0,37,39,41,0,44,46,0,48,50]},
80: {"supply":120, "price":[36,0,38,40,42,0,45,47,0,49]},
81: {"supply":100, "price":[39,0,37,40,0,44,46,48,0,50]},
82: {"supply":105, "price":[0,36,38,0,42,45,0,47,49,51]},
83: {"supply":110, "price":[38,0,41,43,0,46,48,50,0,52]},
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85: {"supply":120, "price":[36,0,38,41,0,45,47,0,49,51]},
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88: {"supply":110, "price":[38,0,40,42,0,45,47,49,0,51]},
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96: {"supply":100, "price":[40,35,0,42,44,0,46,48,50,0]},
97: {"supply":105, "price":[0,36,38,0,41,43,45,0,47,49]},
98: {"supply":110, "price":[38,0,40,42,0,45,47,49,0,51]},
99: {"supply":115, "price":[0,37,39,41,0,44,46,0,48,50]},
100: {"supply":120, "price":[36,0,38,40,42,0,45,47,0,49]}
}

buyers = {
1: {"demand": 90, "price": [75, 76, 0, 78, 79, 0, 81, 82, 83, 0]},
2: {"demand": 100, "price": [0, 68, 70, 72, 0, 74, 75, 0, 77, 78]},
3: {"demand": 85, "price": [0, 66, 68, 0, 70, 71, 72, 73, 74, 0]},
4: {"demand": 95, "price": [80, 0, 81, 82, 83, 0, 85, 86, 0, 87]},
5: {"demand": 100, "price": [0, 70, 72, 73, 74, 75, 0, 77, 78, 0]},
6: {"demand": 110, "price": [68, 69, 0, 71, 72, 73, 0, 75, 76, 0]},
7: {"demand": 78, "price": [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]},
8: {"demand": 105, "price": [82, 83, 0, 84, 0, 85, 86, 0, 87, 88]},
9: {"demand": 115, "price": [0, 67, 69, 70, 71, 0, 72, 73, 0, 74]},
10: {"demand": 98, "price": [75, 0, 76, 77, 0, 78, 79, 80, 0, 81]},
11: {"demand": 120, "price": [0, 70, 72, 0, 74, 75, 76, 77, 78, 0]},
12: {"demand": 86, "price": [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]},
13: {"demand": 92, "price": [0, 65, 67, 0, 69, 70, 71, 72, 73, 0]},
14: {"demand": 88, "price": [76, 77, 78, 0, 79, 80, 81, 0, 82, 83]},
15: {"demand": 113, "price": [0, 72, 73, 74, 0, 75, 76, 77, 0, 78]},
16: {"demand": 109, "price": [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]},
17: {"demand": 91, "price": [0, 69, 70, 0, 72, 73, 74, 75, 0, 76]},
18: {"demand": 104, "price": [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]},
19: {"demand": 97, "price": [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]},
20: {"demand": 118, "price": [0, 66, 68, 69, 0, 70, 71, 72, 73, 0]},
21: {"demand": 99, "price": [83, 0, 84, 85, 0, 86, 87, 88, 0, 89]},
22: {"demand": 84, "price": [0, 70, 72, 73, 0, 74, 75, 76, 0, 77]},
23: {"demand": 110, "price": [80, 0, 81, 82, 83, 0, 84, 85, 86, 0]},
24: {"demand": 93, "price": [0, 69, 70, 0, 71, 72, 73, 74, 75, 0]},
25: {"demand": 102, "price": [77, 78, 0, 79, 0, 80, 81, 82, 0, 83]},
26: {"demand": 108, "price": [0, 73, 74, 75, 0, 76, 77, 0, 78, 79]},
27: {"demand": 96, "price": [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]},
28: {"demand": 87, "price": [0, 65, 67, 68, 0, 70, 71, 72, 0, 73]},
29: {"demand": 116, "price": [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]},
30: {"demand": 90, "price": [0, 68, 69, 0, 70, 71, 72, 73, 0, 74]},
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32: {"demand": 92, "price": [0, 70, 72, 73, 74, 0, 75, 76, 0, 77]},

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33: {"demand": 106, "price": [78, 79, 0, 80, 81, 0, 82, 83, 0, 84]},
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37: {"demand": 119, "price": [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]},
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39: {"demand": 112, "price": [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]},
40: {"demand": 94, "price": [0, 68, 69, 70, 0, 71, 72, 73, 74, 0]},
41: {"demand": 107, "price": [83, 0, 84, 85, 0, 86, 87, 88, 89, 0]},
42: {"demand": 103, "price": [0, 67, 69, 70, 71, 0, 72, 73, 0, 74]},
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56: {"demand": 80, "price": [69, 0, 71, 73, 0, 76, 78, 0, 74, 77]},
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58: {"demand": 90, "price": [70, 0, 72, 74, 0, 76, 78, 80, 0, 75]},
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71: {"demand": 80, "price": [0, 66, 68, 70, 0, 72, 74, 0, 76, 78]},
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82: {"demand": 85, "price": [69, 0, 71, 73, 0, 75, 77, 79, 0, 72]},
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84: {"demand": 95, "price": [70, 0, 72, 0, 74, 76, 0, 78, 80, 0]},
85: {"demand": 100, "price": [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]},
86: {"demand": 80, "price": [71, 0, 73, 75, 0, 77, 79, 0, 72, 74]},
87: {"demand": 85, "price": [0, 69, 71, 0, 73, 75, 77, 0, 79, 81]},
88: {"demand": 90, "price": [70, 0, 72, 74, 0, 76, 78, 0, 75, 77]},
89: {"demand": 95, "price": [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]},
90: {"demand": 100, "price": [71, 0, 73, 0, 75, 77, 0, 79, 81, 0]},
91: {"demand": 80, "price": [0, 66, 68, 70, 0, 72, 74, 0, 76, 78]},
92: {"demand": 85, "price": [69, 0, 71, 73, 0, 75, 77, 79, 0, 72]},

```



```

93: {"demand":90, "price":[0,67,69,71,0,73,75,0,77,79]},
94: {"demand":95, "price":[70,0,72,0,74,76,0,78,80,0]},
95: {"demand":100, "price":[0,68,70,72,0,74,76,78,0,80]},
96: {"demand":80, "price":[71,0,73,75,0,77,79,0,72,74]},
97: {"demand":85, "price":[0,69,71,0,73,75,77,0,79,81]},
98: {"demand":90, "price":[70,0,72,74,0,76,78,0,75,77]},
99: {"demand":95, "price":[0,68,70,72,0,74,76,78,0,80]},
100: {"demand":100, "price":[71,0,73,0,75,77,0,79,81,0]}
}

transporters = {
  1: {"cap": {1:120,2:100,3:90,4:110,5:100,6:130,7:120,8:140,9:100,10:120}, "c
  2: {"cap": {1:80,2:70,3:60,4:90,5:80,6:100,7:85,8:95,9:70,10:80}, "cost":5},
  3: {"cap": {1:100,2:120,3:110,4:130,5:120,6:140,7:135,8:125,9:120,10:115}, "
  4: {"cap": {1:90,2:100,3:80,4:110,5:120,6:130,7:120,8:140,9:110,10:120}, "co
  5: {"cap": {1:70,2:60,3:80,4:75,5:85,6:90,7:95,8:80,9:85,10:90}, "cost":3},
  6: {"cap": {1:100,2:120,3:130,4:110,5:125,6:140,7:135,8:120,9:125,10:130}, "
  7: {"cap": {1:85,2:90,3:95,4:100,5:105,6:110,7:115,8:120,9:125,10:130}, "cos
  8: {"cap": {1:60,2:65,3:70,4:75,5:80,6:85,7:90,8:95,9:100,10:105}, "cost":3}
  9: {"cap": {1:110,2:120,3:130,4:125,5:115,6:140,7:135,8:145,9:130,10:120}, "
  10: {"cap": {1:100,2:105,3:110,4:115,5:120,6:125,7:130,8:135,9:140,10:145},
  11: {"cap": {1:75,2:80,3:85,4:90,5:95,6:100,7:105,8:110,9:115,10:120}, "cost
  12: {"cap": {1:90,2:100,3:110,4:120,5:130,6:140,7:120,8:110,9:100,10:90}, "c
  13: {"cap": {1:60,2:70,3:80,4:90,5:100,6:110,7:120,8:130,9:140,10:150}, "cos
  14: {"cap": {1:130,2:125,3:120,4:115,5:110,6:105,7:100,8:95,9:90,10:85}, "co
  15: {"cap": {1:100,2:95,3:90,4:85,5:80,6:75,7:70,8:65,9:60,10:55}, "cost":3}
  16: {"cap": {1:90,2:100,3:110,4:120,5:130,6:140,7:135,8:125,9:115,10:105}, "
  17: {"cap": {1:80,2:85,3:90,4:95,5:100,6:105,7:110,8:115,9:120,10:125}, "cos
  18: {"cap": {1:100,2:110,3:120,4:130,5:140,6:150,7:135,8:125,9:115,10:105},
  19: {"cap": {1:95,2:90,3:85,4:100,5:110,6:115,7:120,8:125,9:130,10:135}, "co
  20: {"cap": {1:70,2:80,3:90,4:100,5:110,6:120,7:130,8:140,9:150,10:160}, "co
  21: {"cap": {1:120,2:115,3:110,4:105,5:100,6:95,7:90,8:85,9:80,10:75}, "cost
  22: {"cap": {1:100,2:105,3:110,4:115,5:120,6:125,7:130,8:135,9:140,10:145},
  23: {"cap": {1:80,2:85,3:90,4:95,5:100,6:105,7:110,8:115,9:120,10:125}, "cos
  24: {"cap": {1:60,2:70,3:80,4:90,5:100,6:110,7:120,8:130,9:140,10:150}, "cos
  25: {"cap": {1:140,2:130,3:120,4:110,5:100,6:90,7:80,8:70,9:60,10:50}, "cost
  26: {"cap": {1:100,2:110,3:120,4:130,5:140,6:150,7:135,8:125,9:115,10:105},
  27: {"cap": {1:90,2:95,3:100,4:105,5:110,6:115,7:120,8:125,9:130,10:135}, "c
  28: {"cap": {1:80,2:85,3:90,4:95,5:100,6:105,7:110,8:115,9:120,10:125}, "cos
  29: {"cap": {1:70,2:75,3:80,4:85,5:90,6:95,7:100,8:105,9:110,10:115}, "cost"
  30: {"cap": {1:110,2:120,3:130,4:125,5:115,6:140,7:135,8:145,9:130,10:120},
  31: {"cap": {1:100,2:80,3:90,4:0,5:110,6:95,7:100,8:0,9:105,10:90}, "cost":5}
  32: {"cap": {1:50,2:60,3:0,4:55,5:65,6:0,7:70,8:75,9:0,10:60}, "cost":4},
  33: {"cap": {1:80,2:0,3:85,4:90,5:95,6:0,7:100,8:105,9:110,10:0}, "cost":6},
  34: {"cap": {1:60,2:65,3:0,4:70,5:75,6:80,7:0,8:85,9:90,10:0}, "cost":5},
  35: {"cap": {1:90,2:95,3:100,4:0,5:105,6:110,7:0,8:115,9:120,10:0}, "cost":7}
  36: {"cap": {1:55,2:60,3:0,4:65,5:70,6:0,7:75,8:80,9:0,10:85}, "cost":4},
  37: {"cap": {1:70,2:0,3:75,4:80,5:85,6:0,7:90,8:95,9:0,10:100}, "cost":5},
  38: {"cap": {1:60,2:65,3:70,4:0,5:75,6:80,7:0,8:85,9:90,10:0}, "cost":6},
  39: {"cap": {1:50,2:0,3:55,4:60,5:65,6:0,7:70,8:75,9:0,10:80}, "cost":5},
  40: {"cap": {1:80,2:85,3:90,4:0,5:95,6:100,7:0,8:105,9:110,10:0}, "cost":7},
  41: {"cap": {1:55,2:60,3:0,4:65,5:70,6:0,7:75,8:80,9:0,10:85}, "cost":4},
  42: {"cap": {1:70,2:0,3:75,4:80,5:85,6:0,7:90,8:95,9:0,10:100}, "cost":5},
  43: {"cap": {1:60,2:65,3:70,4:0,5:75,6:80,7:0,8:85,9:90,10:0}, "cost":6},
  44: {"cap": {1:50,2:0,3:55,4:60,5:65,6:0,7:70,8:75,9:0,10:80}, "cost":5},
  45: {"cap": {1:80,2:85,3:90,4:0,5:95,6:100,7:0,8:105,9:110,10:0}, "cost":7},
  46: {"cap": {1:55,2:60,3:0,4:65,5:70,6:0,7:75,8:80,9:0,10:85}, "cost":4},
  47: {"cap": {1:70,2:0,3:75,4:80,5:85,6:0,7:90,8:95,9:0,10:100}, "cost":5},
  48: {"cap": {1:60,2:65,3:70,4:0,5:75,6:80,7:0,8:85,9:90,10:0}, "cost":6},
  49: {"cap": {1:50,2:0,3:55,4:60,5:65,6:0,7:70,8:75,9:0,10:80}, "cost":5},

```

```

50: {"cap":{1:80,2:85,3:90,4:0,5:95,6:100,7:0,8:105,9:110,10:0}, "cost":7},
}

T = [1,2,3,4,5,6,7,8,9,10]

```

```

In [7]: # -----
# M 계산
# -----
M = {}
for i in sellers:
    for j in buyers:
        for k in transporters:
            for t in T:
                cap_k_t = transporters[k]["cap"][t]
                M[(i,j,k,t)] = min(sellers[i]["supply"], buyers[j]["demand"], ca

# -----
# valid_tuple 생성
# -----
valid_tuple = []
for (i,j,k,t), mval in M.items():
    net_unit = buyers[j]["price"][t-1] - sellers[i]["price"][t-1] - transporters
    if net_unit >= 0 and (buyers[j]["price"][t-1] > 0) and (sellers[i]["price"]
        valid_tuple.append((int(i), int(j), int(k), int(t)))
valid_tuple = list(set(valid_tuple))
valid_tuple = sorted(valid_tuple, key=lambda x: x[3])

```

```

In [8]: # -----
# 모델 생성
# -----
import gurobipy as gp
from gurobipy import GRB

model = gp.Model("5PL")

# -----
# 변수 정의
# -----
# 기본 변수
L = model.addVars(T, lb=0.0, name="L")
y = model.addVars(valid_tuple, lb=0.0, name="y")
z = model.addVars(valid_tuple, vtype=GRB.BINARY, name="z")

# MA용
p = model.addVars(T, lb=-GRB.INFINITY, name="p")
p_est = model.addVars(T, lb=-GRB.INFINITY, name="p_est")
E = model.addVars(T, lb=0.0, name="E")
B = model.addVars(T, vtype=GRB.BINARY, name="B")
M_val = 1e6

# -----
# 목적함수
# -----
# E: MA 추정치와 실제값 차이
# L: 거래 참여자 간 최소 이윤
obj = gp.quicksum((buyers[j]["price"][t-1] - sellers[i]["price"][t-1] - transpor
    for (i,j,k,t) in valid_tuple) - 0.01 * gp.quicksum(E[t] for t
model.setObjective(obj, GRB.MAXIMIZE)

```

```

# -----
# 제약식
# -----
for (i,j,k,t) in valid_tuple:
    mval = M[(i,j,k,t)]

    # 균형 가격 제약 (거래 활성화시 이윤 보장)
    model.addConstr(p[t] >= sellers[i]["price"][t-1] - mval*(1 - z[i,j,k,t]), name=f"p_lower_{i}_{j}_{k}_{t}")
    model.addConstr(p[t] + transporters[k]["cost"] <= buyers[j]["price"][t-1] + mval*(1 - z[i,j,k,t]), name=f"p_upper_{i}_{j}_{k}_{t}")

    # L[t] 제약 (최소 이윤)
    model.addConstr(L[t] <= buyers[j]["price"][t-1] - p[t] - transporters[k]["cost"], name=f"L_lower_{i}_{j}_{k}_{t}")
    model.addConstr(L[t] <= p[t] - sellers[i]["price"][t-1] + mval*(1 - z[i,j,k,t]), name=f"L_upper_{i}_{j}_{k}_{t}")

# z=0 (거래X) 시점에선 L[t] = 0, p[t] = seller, buyer price 평균
for t in T:
    model.addConstr(L[t] <= M_val * gp.quicksum(z[i,j,k,t] for (i,j,k,tt) in valid_tuple if tt == t), name=f"L_max_{t}")
    sum_z = gp.quicksum(z[i,j,k,t] for (i,j,k,tt) in valid_tuple if tt == t)
    buyer_vals = [buyers[j]["price"][t-1] - transporters[k]["cost"] for (i,j,k,tt) in valid_tuple if tt == t]
    seller_vals = [sellers[i]["price"][t-1] for (i,j,k,tt) in valid_tuple if tt == t]
    if buyer_vals and seller_vals:
        p_mid = 0.5 * (sum(buyer_vals)/len(buyer_vals) + sum(seller_vals)/len(seller_vals))
    else:
        p_mid = p[t-1] if t > 0 else 0 # 첫 시점이면 초기값 0

    model.addConstr(p[t] >= p_mid - M_val * sum_z, name=f"p_lower_{t}")
    model.addConstr(p[t] <= p_mid + M_val * sum_z, name=f"p_upper_{t}")

    if t > 1:
        model.addConstr(p[t] <= p[t-1] + M_val * sum_z, name=f"p_upper_prev_{t}")
        model.addConstr(p[t] >= p[t-1] - M_val * sum_z, name=f"p_lower_prev_{t}")

# y <= M * z
for (i,j,k,t) in valid_tuple:
    model.addConstr(y[i,j,k,t] <= M[(i,j,k,t)] * z[i,j,k,t], name=f"act_{i}_{j}_{k}_{t}")

# seller supply
for i in sellers:
    model.addConstr(gp.quicksum(y[i,j,k,t] for (i,j,k,t) in valid_tuple if ii==i), name=f"supply_{i}")

# buyer demand
for j in buyers:
    model.addConstr(gp.quicksum(y[i,j,k,t] for (i,j,k,t) in valid_tuple if jj==j), name=f"demand_{j}")

# transporter capacity
for k in transporters:
    for t in T:
        model.addConstr(gp.quicksum(y[i,j,k,t] for (i,j,k,t) in valid_tuple if kk==k), name=f"cap_{k}_{t}")

# 라이더는 한 시점당 한번만 운송 가능
for k in transporters:
    for t in T:
        # z[i,j,k,t]의 합 <= 1
        model.addConstr(

```

```

        gp.quicksum(z[i,j,k,t] for (i,j,kk,tt) in valid_tuple if kk==k and t
        name=f"one_trip_per_time_{k}_{t}"
    )

# -----
# MA constraints (이동평균)
# -----
T_list = list(T)
alpha = 0.3
for idx, t in enumerate(T_list):
    if idx < 2:
        model.addConstr(p_est[t] == p[t], name="EMA_init")
    else:
        model.addConstr(p_est[t] == alpha * p[t-1] + (1-alpha) * p_est[t-1])

#  $0.9 * p\_est[t] \leq p[t] \leq 1.1 * p\_est[t]$  일시  $B[t] = 0 \rightarrow E[t] = 0 \rightarrow$  목적함수 영향
# 범위 밖일시  $E[t] \geq |p[t] - p\_est[t]| \rightarrow E[t]$  최소화 위해  $p\_est$ 와 가까워지도록  $p$ 
for t in T:
    model.addConstr(p[t] - 0.9*p_est[t] >= -M_val*(B[t]))
    model.addConstr(1.1*p_est[t] - p[t] >= -M_val*(B[t]))

    model.addConstr(E[t] <= M_val*B[t])
    model.addConstr(E[t] >= p[t] - p_est[t] - M_val*(1 - B[t]))
    model.addConstr(E[t] >= -(p[t] - p_est[t]) - M_val*(1 - B[t]))

model.Params.InfUnbdInfo = 1

# Solve
# -----
model.setParam('OutputFlag', 1)
model.setParam('TimeLimit', 3600)
model.optimize()

```

```

Set parameter Username
Set parameter LicenseID to value 2788128
Academic license - for non-commercial use only - expires 2027-03-06
Set parameter InfUnbdInfo to value 1
Set parameter OutputFlag to value 1
Set parameter TimeLimit to value 3600
Gurobi Optimizer version 13.0.1 build v13.0.1rc0 (win64 - Windows 11+.0 (26200.
2))

```

```

CPU model: AMD Ryzen 9 5900X 12-Core Processor, instruction set [SSE2|AVX|AVX2]
Thread count: 12 physical cores, 24 logical processors, using up to 24 threads

```

Non-default parameters:

TimeLimit 3600

InfUnbdInfo 1

Optimize a model with 10970888 rows, 4387882 columns and 45782490 nonzeros (Max)

Model fingerprint: 0x1c2a0393

Model has 2193918 linear objective coefficients

Variable types: 2193956 continuous, 2193926 integer (2193926 binary)

Coefficient statistics:

```

Matrix range      [3e-01, 1e+06]
Objective range   [1e-02, 5e+01]
Bounds range      [1e+00, 1e+00]
RHS range         [1e+00, 1e+06]

```

Found heuristic solution: objective 16464.622152

```

Presolve removed 136 rows and 18 columns (presolve time = 5s)...
Presolve removed 149 rows and 25 columns (presolve time = 11s)...
Presolve removed 149 rows and 25 columns (presolve time = 15s)...
Presolve removed 149 rows and 25 columns (presolve time = 20s)...
Presolve removed 149 rows and 25 columns (presolve time = 25s)...
Presolve removed 149 rows and 25 columns (presolve time = 30s)...
Presolve removed 149 rows and 25 columns (presolve time = 35s)...
Presolve removed 149 rows and 25 columns (presolve time = 40s)...
Presolve removed 149 rows and 25 columns (presolve time = 45s)...
Presolve removed 149 rows and 25 columns (presolve time = 50s)...
Presolve removed 149 rows and 25 columns (presolve time = 56s)...
Presolve removed 149 rows and 25 columns (presolve time = 61s)...
Presolve removed 149 rows and 25 columns (presolve time = 66s)...
Presolve removed 149 rows and 25 columns (presolve time = 70s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 76s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 80s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 85s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 90s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 96s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 100s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 106s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 110s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 115s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 120s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 125s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 130s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 135s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 140s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 145s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 150s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 159s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 160s)...
Presolve removed 4387099 rows and 25 columns (presolve time = 165s)...

```

Presolve removed 4387099 rows and 25 columns (presolve time = 170s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 175s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 180s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 186s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 191s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 197s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 202s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 205s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 210s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 215s)...
 Presolve removed 4387099 rows and 25 columns (presolve time = 221s)...
 Presolve removed 4387089 rows and 15 columns
 Presolve time: 224.38s
 Presolved: 6583799 rows, 4387867 columns, 41196761 nonzeros
 Variable types: 2193933 continuous, 2193934 integer (2193924 binary)
 Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
 Showing barrier log only...

Root barrier log...

Elapsed ordering time = 6s
 Elapsed ordering time = 9s
 Elapsed ordering time = 12s
 Elapsed ordering time = 15s
 Ordering time: 17.72s

Barrier statistics:

Dense cols : 801
 AA' NZ : 5.667e+07
 Factor NZ : 8.769e+07 (roughly 5.0 GB of memory)
 Factor Ops : 3.439e+09 (less than 1 second per iteration)
 Threads : 10

Iter	Objective		Residual		Compl	Time
	Primal	Dual	Primal	Dual		
0	2.24073459e+13	9.76332465e+08	1.91e+09	8.94e+01	7.01e+06	305s
1	1.12815325e+13	2.10222772e+08	9.60e+08	6.94e+01	3.01e+06	310s
2	1.11603758e+13	1.85756059e+08	9.50e+08	5.58e+01	2.87e+06	316s
3	1.08965827e+13	2.00083356e+08	9.28e+08	5.02e+01	2.75e+06	323s
4	1.01564846e+13	2.75415822e+08	8.65e+08	4.14e+01	2.48e+06	328s
5	9.73227049e+12	3.24727562e+08	8.28e+08	3.63e+01	2.34e+06	332s
6	9.05205082e+12	4.77167428e+08	7.70e+08	2.65e+01	2.06e+06	337s
7	8.36383390e+12	7.75032390e+08	7.12e+08	1.49e+01	1.79e+06	344s
8	7.20454933e+12	1.23212682e+09	6.13e+08	3.01e+00	1.49e+06	351s
9	4.74149848e+12	1.44777954e+09	4.03e+08	6.05e-07	9.89e+05	359s
10	5.66260526e+09	1.44630439e+09	4.83e+05	6.18e-07	1.28e+03	366s
11	2.30559885e+09	8.26921553e+08	1.97e+05	1.05e-06	5.08e+02	373s
12	1.88657154e+08	4.23298798e+08	1.61e+04	6.54e-07	6.29e+01	379s
13	7.57947526e+07	2.24687372e+08	6.44e+03	1.57e-07	2.84e+01	383s
14	3.73571811e+07	1.13962269e+08	3.16e+03	9.74e-08	1.41e+01	387s
15	1.15559603e+07	7.04801532e+07	9.61e+02	6.08e-08	6.60e+00	391s
16	3.60288191e+06	4.83127383e+07	2.84e+02	3.11e-08	3.72e+00	395s
17	1.42316382e+06	3.43005247e+07	9.87e+01	2.40e-08	2.41e+00	400s
18	8.37070553e+05	2.68767249e+07	4.88e+01	6.29e-09	1.82e+00	404s
19	5.08613045e+05	2.05746995e+07	2.09e+01	8.50e-09	1.36e+00	409s
20	3.97366603e+05	1.53575181e+07	1.15e+01	1.42e-08	1.00e+00	413s
21	2.76524611e+05	1.18683164e+07	1.26e+00	1.50e-08	7.57e-01	417s
22	2.73367757e+05	4.15470639e+06	9.09e-01	5.06e-09	2.54e-01	421s
23	2.73126561e+05	3.25491743e+06	7.95e-01	3.31e-09	1.95e-01	426s
24	2.73536463e+05	2.92097238e+06	6.21e-01	2.03e-09	1.73e-01	430s

25	2.75243248e+05	2.09430561e+06	4.18e-01	2.24e-09	1.19e-01	434s
26	2.77236905e+05	1.24905798e+06	2.91e-01	9.79e-10	6.36e-02	440s
27	2.77553048e+05	1.15054561e+06	2.83e-01	8.11e-10	5.71e-02	447s
28	2.78755052e+05	9.78347090e+05	2.55e-01	6.13e-10	4.58e-02	456s
29	2.83677586e+05	7.17863168e+05	1.60e-01	4.06e-10	2.84e-02	464s
30	2.90906847e+05	6.10149552e+05	7.64e-02	3.33e-10	2.09e-02	471s
31	2.93676995e+05	5.18240952e+05	5.90e-02	2.12e-10	1.47e-02	476s
32	2.95716056e+05	4.54290315e+05	4.71e-02	1.89e-10	1.04e-02	484s
33	3.00580553e+05	4.15240446e+05	2.25e-02	1.24e-10	7.49e-03	491s
34	3.01715480e+05	3.79928008e+05	1.76e-02	6.37e-11	5.11e-03	500s
35	3.03926380e+05	3.63332796e+05	8.95e-03	6.01e-11	3.88e-03	509s
36	3.05319650e+05	3.46196779e+05	3.69e-03	5.05e-11	2.66e-03	518s
37	3.05535964e+05	3.41736430e+05	3.01e-03	4.34e-11	2.36e-03	526s
38	3.05621353e+05	3.23829333e+05	2.75e-03	2.52e-11	1.19e-03	531s
39	3.06065858e+05	3.17432628e+05	1.35e-03	1.67e-11	7.41e-04	537s
40	3.06350988e+05	3.10580515e+05	4.23e-04	5.39e-12	2.76e-04	543s
41	3.06417531e+05	3.08282713e+05	2.66e-04	7.15e-12	1.22e-04	547s
42	3.06497770e+05	3.07559025e+05	9.06e-05	2.64e-12	6.92e-05	551s
43	3.06531436e+05	3.06939488e+05	1.76e-05	1.82e-13	2.66e-05	556s
44	3.06534311e+05	3.06881910e+05	1.39e-05	1.54e-13	2.26e-05	561s
45	3.06537359e+05	3.06715364e+05	8.63e-06	1.14e-13	1.16e-05	566s
46	3.06542086e+05	3.06619584e+05	1.91e-06	1.21e-13	5.05e-06	570s
47	3.06543886e+05	3.06563193e+05	1.38e-08	6.44e-13	1.26e-06	575s
48	3.06544218e+05	3.06562490e+05	4.82e-09	6.49e-13	1.19e-06	581s
49	3.06544887e+05	3.06559556e+05	3.16e-09	4.14e-13	9.55e-07	587s
50	3.06545385e+05	3.06557529e+05	2.76e-09	5.72e-13	7.91e-07	591s
51	3.06546188e+05	3.06557895e+05	2.73e-09	5.76e-13	7.62e-07	597s
52	3.06548214e+05	3.06557151e+05	2.32e-09	6.02e-13	5.82e-07	604s
53	3.06548442e+05	3.06554649e+05	2.35e-09	9.60e-13	4.04e-07	612s
54	3.06548912e+05	3.06554361e+05	2.24e-09	9.95e-13	3.55e-07	622s
55	3.06549444e+05	3.06553725e+05	2.05e-09	9.47e-13	2.79e-07	633s
56	3.06549820e+05	3.06552544e+05	1.76e-09	1.99e-12	1.77e-07	641s
57	3.06549968e+05	3.06551662e+05	1.48e-09	1.70e-12	1.10e-07	652s
58	3.06549978e+05	3.06551493e+05	1.48e-09	1.22e-12	9.86e-08	663s
59	3.06550022e+05	3.06551302e+05	1.27e-09	7.55e-13	8.33e-08	674s
60	3.06550110e+05	3.06551045e+05	2.99e-09	3.11e-13	6.09e-08	684s
61	3.06550115e+05	3.06550833e+05	2.95e-09	1.32e-13	4.68e-08	692s
62	3.06550147e+05	3.06550438e+05	2.15e-09	1.28e-13	1.89e-08	704s
63	3.06550165e+05	3.06550333e+05	1.67e-09	7.16e-13	1.09e-08	713s
64	3.06550174e+05	3.06550316e+05	2.43e-09	2.07e-13	9.23e-09	723s
65	3.06550183e+05	3.06550291e+05	1.26e-09	2.32e-13	7.06e-09	730s
66	3.06550184e+05	3.06550278e+05	1.20e-09	1.72e-13	6.08e-09	738s
67	3.06550191e+05	3.06550254e+05	1.05e-09	1.14e-13	4.08e-09	746s
68	3.06550220e+05	3.06550237e+05	3.77e-09	1.42e-13	1.12e-09	753s
69	3.06550232e+05	3.06550233e+05	5.14e-10	2.81e-13	6.50e-11	760s

Barrier solved model in 69 iterations and 760.57 seconds (729.09 work units)
Optimal objective 3.06550232e+05

Root crossover log...

21154 DPushes remaining with DInf 0.0000000e+00	771s
2865 DPushes remaining with DInf 0.0000000e+00	778s
1473 DPushes remaining with DInf 0.0000000e+00	790s
0 DPushes remaining with DInf 0.0000000e+00	796s
1959459 PPushes remaining with PInf 0.0000000e+00	797s
804021 PPushes remaining with PInf 0.0000000e+00	802s
788952 PPushes remaining with PInf 0.0000000e+00	808s

```

778264 PPushes remaining with PInf 0.000000e+00      814s
759860 PPushes remaining with PInf 0.000000e+00      821s
655683 PPushes remaining with PInf 0.000000e+00      826s
635510 PPushes remaining with PInf 0.000000e+00      831s
Warning: 1 variables dropped from basis

```

Restart crossover...

```

1964287 variables added to crossover basis      836s
6541471 variables added to crossover basis      841s

21163 DPushes remaining with DInf 0.000000e+00      842s
2872 DPushes remaining with DInf 0.000000e+00      849s
1479 DPushes remaining with DInf 0.000000e+00      856s
0 DPushes remaining with DInf 0.000000e+00      864s

1959459 PPushes remaining with PInf 0.000000e+00      865s
215524 PPushes remaining with PInf 0.000000e+00      871s
195325 PPushes remaining with PInf 0.000000e+00      876s
107688 PPushes remaining with PInf 0.000000e+00      886s
99518 PPushes remaining with PInf 0.000000e+00      891s
13641 PPushes remaining with PInf 0.000000e+00      923s
Warning: 1 variables dropped from basis
11922 PPushes remaining with PInf 0.000000e+00      925s
Warning: 2 variables dropped from basis
917 PPushes remaining with PInf 3.3008931e+02      931s
0 PPushes remaining with PInf 1.3037226e+02      933s

Push phase complete: Pinf 1.3037226e+02, Dinf 8.9630317e+01 934s

```

Root simplex log...

Iteration	Objective	Primal Inf.	Dual Inf.	Time
1979629	3.0655023e+05	0.000000e+00	8.963032e+01	936s
1979729	3.0655023e+05	0.000000e+00	1.455196e+01	978s
1980538	3.0655023e+05	9.606460e+02	0.000000e+00	997s
1981340	3.0655023e+05	2.019280e+02	0.000000e+00	1011s
1982142	3.0655023e+05	1.675698e+02	0.000000e+00	1035s
1982932	3.0655023e+05	5.934998e+02	0.000000e+00	1069s
1983372	3.0655023e+05	1.312551e+02	0.000000e+00	1099s
1983912	3.0655023e+05	2.218503e+02	0.000000e+00	1129s
1984702	3.0655023e+05	5.468843e+02	0.000000e+00	1163s
1985382	3.0655023e+05	4.044278e+02	0.000000e+00	1186s
1985822	3.0655023e+05	2.600530e+02	0.000000e+00	1207s
1986472	3.0655023e+05	4.061226e+02	0.000000e+00	1231s
1987182	3.0655023e+05	3.778600e+02	0.000000e+00	1254s
1987984	3.0655023e+05	2.407355e+02	0.000000e+00	1285s
1988786	3.0655023e+05	1.104774e+03	0.000000e+00	1316s
1989588	3.0655023e+05	2.269099e+03	0.000000e+00	1336s
1990390	3.0655023e+05	5.139802e+02	0.000000e+00	1361s
1990960	3.0655023e+05	1.068877e+03	0.000000e+00	1384s
1991480	3.0655023e+05	7.789594e+02	0.000000e+00	1405s
1991830	3.0655023e+05	9.052062e+02	0.000000e+00	1427s
1992270	3.0655023e+05	6.419176e+02	0.000000e+00	1451s
1993072	3.0655023e+05	3.723293e+02	0.000000e+00	1477s
1993702	3.0655023e+05	4.783454e+02	0.000000e+00	1499s
1994052	3.0655023e+05	1.026291e+03	0.000000e+00	1520s
1994622	3.0655023e+05	3.209296e+02	0.000000e+00	1541s
1995022	3.0655023e+05	2.504489e+02	0.000000e+00	1559s

```

1995462    3.0655023e+05    8.666704e+02    0.000000e+00    1580s
1996042    3.0655023e+05    3.888156e+02    0.000000e+00    1601s
1996502    3.0655023e+05    6.513649e+02    0.000000e+00    1621s
1997012    3.0655023e+05    1.693562e+02    0.000000e+00    1640s
1997442    3.0655023e+05    1.101331e+02    0.000000e+00    1660s
1997762    3.0655023e+05    1.713734e+02    0.000000e+00    1680s
1998092    3.0655023e+05    8.091028e+01    0.000000e+00    1704s
1998572    3.0655023e+05    7.890399e+01    0.000000e+00    1739s
1998832    3.0655023e+05    9.380436e+01    0.000000e+00    1758s
1999112    3.0655023e+05    5.698973e+01    0.000000e+00    1776s
1999432    3.0655023e+05    6.291135e+01    0.000000e+00    1796s
1999712    3.0655023e+05    2.293255e+02    0.000000e+00    1817s
2000212    3.0655023e+05    1.975192e+02    0.000000e+00    1856s
2000572    3.0655023e+05    2.181417e+02    0.000000e+00    1884s
2000862    3.0655022e+05    2.749786e+02    0.000000e+00    1903s
2001132    3.0655022e+05    2.949464e+02    0.000000e+00    1923s
2001422    3.0655022e+05    1.121858e+02    0.000000e+00    1949s
2001712    3.0655022e+05    2.881671e+00    0.000000e+00    1970s
2001962    3.0655022e+05    5.850023e-01    0.000000e+00    1990s
2002124    3.0655022e+05    0.000000e+00    0.000000e+00    2006s
2002235    3.0655023e+05    0.000000e+00    0.000000e+00    2036s

```

Crossover time: 1274.78 seconds (3618.95 work units)

Waiting for other threads to finish... 2226s

Waiting for other threads to finish... 2416s

Waiting for other threads to finish... 2607s

Waiting for other threads to finish... 2797s

Waiting for other threads to finish... 2985s

Waiting for other threads to finish... 3174s

Waiting for other threads to finish... 3363s

Waiting for other threads to finish... 3552s

Concurrent spin time: 1563.94s (can be avoided by choosing Method=3)

Solved with barrier

Root relaxation: objective 3.065502e+05, 2002235 iterations, 3356.15 seconds (388 5.27 work units)

Nodes		Current Node			Objective Bounds			Work	
Expl	Unexpl	Obj	Depth	IntInf	Incumbent	BestBd	Gap	It/Node	Time
0	0	306550.232	0	1333	16464.6222	306550.232	1762%	-	3600s

Explored 1 nodes (2002235 simplex iterations) in 3602.50 seconds (4348.13 work units)

Thread count was 24 (of 24 available processors)

Solution count 2: 16464.6 16464.6

Time limit reached

Best objective 1.646462215205e+04, best bound 3.065502319189e+05, gap 1761.8723%

```

In [9]: import pickle
        solution = {
            "L": {t: L[t].X for t in T},
            "y": {(i,j,k,t): y[i,j,k,t].X for (i,j,k,t) in valid_tuple},
            "z": {(i,j,k,t): z[i,j,k,t].X for (i,j,k,t) in valid_tuple},
            "p": {t: p[t].X for t in T},
            "p_est": {t: p_est[t].X for t in T},
            "E": {t: E[t].X for t in T},
            "B": {t: B[t].X for t in T},

```

```
}

with open("5PL_large.pkl", "wb") as f:
    pickle.dump(solution, f)
```

```
In [10]: # -----
# 결과 출력
# -----
with open("5PL_large.pkl", "rb") as f:
    solution = pickle.load(f)

y_sol = solution["y"]
p_sol = solution["p"]
p_est_sol = solution["p_est"]

# -----
seller_receive = {i: [] for i in sellers}      # (t, qty, price_per_unit)
buyer_pay = {j: [] for j in buyers}           # (t, qty, price_per_unit)
transporter_receive = {k: [] for k in transporters} # (t, qty, price_per_unit)

prev = 0
for (i,j,k,t) in valid_tuple:
    q = y_sol[i,j,k,t]
    if q > 1e-6:
        pt_val = p_sol[t]
        if t > prev:
            print()
            print(f"t={t}, p = {pt_val:.1f}, p_est = {p_est_sol[t]:.1f}")
            prev = t
        cost_val = transporters[k]["cost"]
        print(f"{i} -({k})-> {j} / qty: {int(q)} / intial price: {int(sellers[i]

        seller_receive[i].append((t, q, pt_val))
        buyer_pay[j].append((t, q, pt_val + cost_val))
        transporter_receive[k].append((t, q, cost_val))

print("\n=== 참여자별 정리 ===")
seller_qty = {i:0.0 for i in sellers}
buyer_qty = {j:0.0 for j in buyers}
transporter_qty = {k:0.0 for k in transporters}

print('sellers')
for i in sellers:
    for (t, q, unit_price) in seller_receive[i]:
        seller_qty[i] += q
    print(i, ':', sellers[i], 'Total qty =', int(seller_qty[i]))
print()
print('buyers')
for i in buyers:
    for (t, q, unit_price) in buyer_pay[i]:
        buyer_qty[i] += q
    print(i, ':', buyers[i], 'Total qty =', int(buyer_qty[i]))
print()
print('transporters')
for i in transporters:
    for (t, q, unit_price) in transporter_receive[i]:
        transporter_qty[i] += q
    print(i, ':', transporters[i], 'Total qty =', int(transporter_qty[i]))
```

```
print()  
  
# 주체별 총액 및 총량 초기화  
seller_total = {i:0.0 for i in sellers}  
buyer_total = {j:0.0 for j in buyers}  
transporter_total = {k:0.0 for k in transporters}
```

t=1, p = 50.0, p_est = 50.0
43 -(6)-> 80 / qty: 100 / intial price: 50 75

t=2, p = 39.0, p_est = 39.0
48 -(23)-> 20 / qty: 85 / intial price: 39 66

t=3, p = 43.0, p_est = 39.0
49 -(42)-> 88 / qty: 75 / intial price: 43 72

t=4, p = 45.3, p_est = 40.2
12 -(49)-> 68 / qty: 60 / intial price: 45 74

t=5, p = 42.0, p_est = 41.7
33 -(22)-> 3 / qty: 85 / intial price: 42 70

t=6, p = 46.0, p_est = 41.8
83 -(25)-> 51 / qty: 80 / intial price: 46 70

t=7, p = 49.0, p_est = 43.1
2 -(46)-> 96 / qty: 75 / intial price: 49 79

t=8, p = 57.0, p_est = 44.9
18 -(37)-> 75 / qty: 95 / intial price: 57 78

t=9, p = 43.0, p_est = 48.5
15 -(50)-> 86 / qty: 80 / intial price: 43 72

t=10, p = 52.0, p_est = 46.8
83 -(14)-> 25 / qty: 30 / intial price: 52 83

=== 참여자별 정리 ===

sellers

1 : {'supply': 124, 'price': [45, 0, 52, 40, 58, 0, 64, 59, 61, 55]} Total qty = 0
2 : {'supply': 97, 'price': [0, 48, 51, 54, 0, 50, 49, 0, 53, 47]} Total qty = 75
3 : {'supply': 142, 'price': [40, 45, 0, 42, 43, 44, 0, 46, 48, 49]} Total qty = 0
4 : {'supply': 120, 'price': [0, 38, 41, 43, 0, 46, 45, 47, 50, 52]} Total qty = 0
5 : {'supply': 133, 'price': [35, 0, 39, 40, 0, 42, 43, 44, 46, 0]} Total qty = 0
6 : {'supply': 106, 'price': [0, 34, 36, 0, 38, 40, 42, 44, 45, 0]} Total qty = 0
7 : {'supply': 128, 'price': [50, 52, 0, 54, 56, 57, 0, 59, 0, 60]} Total qty = 0
8 : {'supply': 114, 'price': [45, 47, 0, 48, 49, 0, 51, 52, 54, 55]} Total qty = 0
9 : {'supply': 136, 'price': [0, 43, 45, 0, 48, 50, 0, 52, 53, 55]} Total qty = 0
10 : {'supply': 89, 'price': [35, 37, 0, 38, 39, 0, 41, 0, 43, 44]} Total qty = 0
11 : {'supply': 141, 'price': [55, 57, 58, 0, 60, 61, 0, 63, 64, 0]} Total qty = 0
12 : {'supply': 115, 'price': [42, 0, 44, 45, 46, 0, 47, 48, 49, 0]} Total qty = 60
13 : {'supply': 124, 'price': [39, 40, 0, 41, 0, 42, 43, 44, 45, 46]} Total qty = 0
14 : {'supply': 132, 'price': [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]} Total qty = 0
15 : {'supply': 107, 'price': [0, 36, 38, 39, 0, 40, 41, 42, 43, 44]} Total qty = 80
16 : {'supply': 120, 'price': [49, 0, 51, 0, 52, 54, 0, 56, 57, 0]} Total qty = 0
17 : {'supply': 103, 'price': [40, 42, 0, 43, 44, 45, 46, 0, 47, 0]} Total qty = 0
18 : {'supply': 129, 'price': [0, 50, 52, 53, 0, 55, 56, 57, 58, 0]} Total qty =

```

95
19 : {'supply': 118, 'price': [37, 38, 0, 39, 40, 41, 42, 0, 43, 44]} Total qty =
0
20 : {'supply': 125, 'price': [0, 45, 46, 47, 48, 0, 49, 50, 51, 0]} Total qty =
0
21 : {'supply': 132, 'price': [55, 0, 57, 50, 59, 0, 60, 61, 0, 63]} Total qty =
0
22 : {'supply': 139, 'price': [40, 41, 0, 42, 43, 0, 44, 45, 0, 46]} Total qty =
0
23 : {'supply': 127, 'price': [0, 48, 49, 50, 51, 52, 0, 53, 54, 0]} Total qty =
0
24 : {'supply': 133, 'price': [0, 37, 38, 39, 40, 41, 42, 0, 43, 0]} Total qty =
0
25 : {'supply': 144, 'price': [42, 0, 43, 44, 0, 46, 47, 48, 49, 0]} Total qty =
0
26 : {'supply': 95, 'price': [0, 50, 52, 53, 54, 0, 55, 56, 57, 0]} Total qty = 0
27 : {'supply': 130, 'price': [39, 0, 40, 41, 42, 43, 0, 44, 45, 0]} Total qty =
0
28 : {'supply': 138, 'price': [0, 47, 48, 0, 50, 51, 52, 0, 53, 54]} Total qty =
0
29 : {'supply': 101, 'price': [36, 37, 38, 0, 39, 0, 40, 41, 0, 42]} Total qty =
0
30 : {'supply': 119, 'price': [0, 44, 45, 46, 47, 48, 0, 49, 50, 0]} Total qty =
0
31 : {'supply': 140, 'price': [51, 0, 52, 53, 54, 0, 55, 56, 57, 0]} Total qty =
0
32 : {'supply': 111, 'price': [0, 42, 43, 0, 44, 45, 46, 47, 0, 48]} Total qty =
0
33 : {'supply': 108, 'price': [39, 0, 40, 41, 42, 43, 0, 44, 45, 46]} Total qty =
85
34 : {'supply': 136, 'price': [0, 49, 50, 51, 52, 0, 53, 54, 0, 55]} Total qty =
0
35 : {'supply': 122, 'price': [46, 0, 48, 49, 0, 50, 51, 0, 52, 53]} Total qty =
0
36 : {'supply': 118, 'price': [0, 35, 36, 37, 0, 38, 39, 40, 41, 0]} Total qty =
0
37 : {'supply': 131, 'price': [40, 41, 42, 0, 43, 44, 0, 45, 46, 0]} Total qty =
0
38 : {'supply': 100, 'price': [0, 45, 47, 0, 48, 49, 50, 0, 51, 52]} Total qty =
0
39 : {'supply': 129, 'price': [37, 0, 38, 39, 40, 0, 41, 42, 43, 0]} Total qty =
0
40 : {'supply': 105, 'price': [0, 53, 54, 0, 55, 56, 57, 0, 58, 59]} Total qty =
0
41 : {'supply': 117, 'price': [44, 0, 45, 46, 47, 0, 48, 49, 50, 0]} Total qty =
0
42 : {'supply': 125, 'price': [0, 38, 39, 40, 41, 0, 42, 43, 0, 44]} Total qty =
0
43 : {'supply': 130, 'price': [50, 0, 52, 0, 53, 54, 0, 55, 56, 57]} Total qty =
100
44 : {'supply': 126, 'price': [0, 42, 43, 44, 0, 45, 46, 0, 47, 48]} Total qty =
0
45 : {'supply': 121, 'price': [37, 38, 0, 39, 0, 40, 41, 42, 43, 0]} Total qty =
0
46 : {'supply': 110, 'price': [0, 47, 48, 49, 50, 0, 51, 52, 0, 53]} Total qty =
0
47 : {'supply': 119, 'price': [45, 0, 46, 47, 48, 0, 49, 50, 51, 0]} Total qty =
0
48 : {'supply': 123, 'price': [0, 39, 40, 41, 0, 42, 43, 44, 0, 45]} Total qty =
85

```



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49 : {'supply': 129, 'price': [41, 0, 43, 44, 0, 45, 46, 47, 0, 48]} Total qty =
75
50 : {'supply': 134, 'price': [0, 49, 50, 51, 0, 52, 53, 54, 55, 0]} Total qty =
0
51 : {'supply': 100, 'price': [40, 35, 38, 0, 42, 45, 50, 0, 55, 48]} Total qty =
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52 : {'supply': 105, 'price': [0, 36, 0, 41, 44, 40, 0, 47, 49, 42]} Total qty =
0
53 : {'supply': 110, 'price': [38, 0, 42, 45, 0, 48, 50, 52, 0, 46]} Total qty =
0
54 : {'supply': 115, 'price': [35, 37, 39, 0, 44, 0, 50, 53, 55, 0]} Total qty =
0
55 : {'supply': 120, 'price': [0, 36, 38, 40, 42, 0, 48, 50, 52, 54]} Total qty =
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56 : {'supply': 100, 'price': [42, 0, 40, 43, 0, 47, 45, 0, 50, 48]} Total qty =
0
57 : {'supply': 105, 'price': [0, 39, 0, 41, 44, 46, 0, 49, 51, 0]} Total qty = 0
58 : {'supply': 110, 'price': [38, 36, 0, 0, 42, 45, 47, 0, 50, 53]} Total qty =
0
59 : {'supply': 115, 'price': [35, 0, 39, 42, 0, 46, 48, 50, 0, 52]} Total qty =
0
60 : {'supply': 120, 'price': [0, 36, 38, 41, 43, 0, 47, 49, 51, 0]} Total qty =
0
61 : {'supply': 100, 'price': [40, 0, 37, 39, 42, 0, 45, 47, 0, 50]} Total qty =
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62 : {'supply': 105, 'price': [0, 36, 38, 0, 41, 44, 0, 46, 48, 50]} Total qty =
0
63 : {'supply': 110, 'price': [39, 0, 42, 45, 0, 47, 50, 52, 0, 49]} Total qty =
0
64 : {'supply': 115, 'price': [0, 37, 39, 41, 0, 44, 46, 48, 0, 51]} Total qty =
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65 : {'supply': 120, 'price': [38, 0, 0, 42, 45, 0, 47, 49, 51, 53]} Total qty =
0
66 : {'supply': 100, 'price': [40, 35, 0, 43, 0, 46, 48, 50, 0, 52]} Total qty =
0
67 : {'supply': 105, 'price': [0, 36, 38, 40, 0, 44, 46, 0, 49, 51]} Total qty =
0
68 : {'supply': 110, 'price': [37, 0, 39, 41, 0, 45, 47, 49, 0, 50]} Total qty =
0
69 : {'supply': 115, 'price': [0, 36, 38, 0, 42, 44, 46, 48, 0, 51]} Total qty =
0
70 : {'supply': 120, 'price': [38, 0, 40, 42, 0, 45, 47, 0, 50, 52]} Total qty =
0
71 : {'supply': 100, 'price': [39, 0, 37, 0, 42, 44, 46, 48, 0, 50]} Total qty =
0
72 : {'supply': 105, 'price': [0, 36, 38, 40, 0, 43, 45, 0, 47, 49]} Total qty =
0
73 : {'supply': 110, 'price': [38, 0, 41, 43, 0, 46, 48, 50, 0, 52]} Total qty =
0
74 : {'supply': 115, 'price': [0, 37, 39, 0, 44, 46, 0, 48, 50, 52]} Total qty =
0
75 : {'supply': 120, 'price': [36, 0, 38, 41, 0, 45, 47, 0, 49, 51]} Total qty =
0
76 : {'supply': 100, 'price': [40, 35, 0, 42, 44, 0, 46, 48, 50, 0]} Total qty =
0
77 : {'supply': 105, 'price': [0, 36, 38, 0, 41, 43, 45, 0, 47, 49]} Total qty =
0
78 : {'supply': 110, 'price': [38, 0, 40, 42, 0, 45, 47, 49, 0, 51]} Total qty =
0
79 : {'supply': 115, 'price': [0, 37, 39, 41, 0, 44, 46, 0, 48, 50]} Total qty =

```

```

0
80 : {'supply': 120, 'price': [36, 0, 38, 40, 42, 0, 45, 47, 0, 49]} Total qty =
0
81 : {'supply': 100, 'price': [39, 0, 37, 40, 0, 44, 46, 48, 0, 50]} Total qty =
0
82 : {'supply': 105, 'price': [0, 36, 38, 0, 42, 45, 0, 47, 49, 51]} Total qty =
0
83 : {'supply': 110, 'price': [38, 0, 41, 43, 0, 46, 48, 50, 0, 52]} Total qty =
110
84 : {'supply': 115, 'price': [0, 37, 39, 0, 44, 46, 48, 0, 50, 52]} Total qty =
0
85 : {'supply': 120, 'price': [36, 0, 38, 41, 0, 45, 47, 0, 49, 51]} Total qty =
0
86 : {'supply': 100, 'price': [40, 0, 37, 39, 0, 44, 46, 48, 50, 0]} Total qty =
0
87 : {'supply': 105, 'price': [0, 36, 38, 40, 0, 43, 45, 0, 47, 49]} Total qty =
0
88 : {'supply': 110, 'price': [38, 0, 40, 42, 0, 45, 47, 49, 0, 51]} Total qty =
0
89 : {'supply': 115, 'price': [0, 37, 39, 41, 0, 44, 46, 0, 48, 50]} Total qty =
0
90 : {'supply': 120, 'price': [36, 0, 38, 40, 42, 0, 45, 47, 0, 49]} Total qty =
0
91 : {'supply': 100, 'price': [39, 0, 37, 0, 42, 44, 46, 48, 0, 50]} Total qty =
0
92 : {'supply': 105, 'price': [0, 36, 38, 40, 0, 43, 45, 0, 47, 49]} Total qty =
0
93 : {'supply': 110, 'price': [38, 0, 41, 43, 0, 46, 48, 50, 0, 52]} Total qty =
0
94 : {'supply': 115, 'price': [0, 37, 39, 0, 44, 46, 0, 48, 50, 52]} Total qty =
0
95 : {'supply': 120, 'price': [36, 0, 38, 41, 0, 45, 47, 0, 49, 51]} Total qty =
0
96 : {'supply': 100, 'price': [40, 35, 0, 42, 44, 0, 46, 48, 50, 0]} Total qty =
0
97 : {'supply': 105, 'price': [0, 36, 38, 0, 41, 43, 45, 0, 47, 49]} Total qty =
0
98 : {'supply': 110, 'price': [38, 0, 40, 42, 0, 45, 47, 49, 0, 51]} Total qty =
0
99 : {'supply': 115, 'price': [0, 37, 39, 41, 0, 44, 46, 0, 48, 50]} Total qty =
0
100 : {'supply': 120, 'price': [36, 0, 38, 40, 42, 0, 45, 47, 0, 49]} Total qty =
0

```

buyers

```

1 : {'demand': 90, 'price': [75, 76, 0, 78, 79, 0, 81, 82, 83, 0]} Total qty = 0
2 : {'demand': 100, 'price': [0, 68, 70, 72, 0, 74, 75, 0, 77, 78]} Total qty = 0
3 : {'demand': 85, 'price': [0, 66, 68, 0, 70, 71, 72, 73, 74, 0]} Total qty = 85
4 : {'demand': 95, 'price': [80, 0, 81, 82, 83, 0, 85, 86, 0, 87]} Total qty = 0
5 : {'demand': 100, 'price': [0, 70, 72, 73, 74, 75, 0, 77, 78, 0]} Total qty = 0
6 : {'demand': 110, 'price': [68, 69, 0, 71, 72, 73, 0, 75, 76, 0]} Total qty = 0
7 : {'demand': 78, 'price': [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]} Total qty = 0
8 : {'demand': 105, 'price': [82, 83, 0, 84, 0, 85, 86, 0, 87, 88]} Total qty = 0
9 : {'demand': 115, 'price': [0, 67, 69, 70, 71, 0, 72, 73, 0, 74]} Total qty = 0
10 : {'demand': 98, 'price': [75, 0, 76, 77, 0, 78, 79, 80, 0, 81]} Total qty = 0
11 : {'demand': 120, 'price': [0, 70, 72, 0, 74, 75, 76, 77, 78, 0]} Total qty =
0
12 : {'demand': 86, 'price': [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]} Total qty = 0
13 : {'demand': 92, 'price': [0, 65, 67, 0, 69, 70, 71, 72, 73, 0]} Total qty = 0
14 : {'demand': 88, 'price': [76, 77, 78, 0, 79, 80, 81, 0, 82, 83]} Total qty =

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0
15 : {'demand': 113, 'price': [0, 72, 73, 74, 0, 75, 76, 77, 0, 78]} Total qty =
0
16 : {'demand': 109, 'price': [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]} Total qty =
0
17 : {'demand': 91, 'price': [0, 69, 70, 0, 72, 73, 74, 75, 0, 76]} Total qty = 0
18 : {'demand': 104, 'price': [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]} Total qty =
0
19 : {'demand': 97, 'price': [0, 74, 75, 76, 0, 77, 78, 79, 0, 80]} Total qty = 0
20 : {'demand': 118, 'price': [0, 66, 68, 69, 0, 70, 71, 72, 73, 0]} Total qty =
85
21 : {'demand': 99, 'price': [83, 0, 84, 85, 0, 86, 87, 88, 0, 89]} Total qty = 0
22 : {'demand': 84, 'price': [0, 70, 72, 73, 0, 74, 75, 76, 0, 77]} Total qty = 0
23 : {'demand': 110, 'price': [80, 0, 81, 82, 83, 0, 84, 85, 86, 0]} Total qty =
0
24 : {'demand': 93, 'price': [0, 69, 70, 0, 71, 72, 73, 74, 75, 0]} Total qty = 0
25 : {'demand': 102, 'price': [77, 78, 0, 79, 0, 80, 81, 82, 0, 83]} Total qty =
30
26 : {'demand': 108, 'price': [0, 73, 74, 75, 0, 76, 77, 0, 78, 79]} Total qty =
0
27 : {'demand': 96, 'price': [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]} Total qty = 0
28 : {'demand': 87, 'price': [0, 65, 67, 68, 0, 70, 71, 72, 0, 73]} Total qty = 0
29 : {'demand': 116, 'price': [79, 0, 80, 81, 82, 0, 83, 84, 0, 85]} Total qty =
0
30 : {'demand': 90, 'price': [0, 68, 69, 0, 70, 71, 72, 73, 0, 74]} Total qty = 0
31 : {'demand': 114, 'price': [84, 0, 85, 86, 0, 87, 88, 89, 0, 90]} Total qty =
0
32 : {'demand': 92, 'price': [0, 70, 72, 73, 74, 0, 75, 76, 0, 77]} Total qty = 0
33 : {'demand': 106, 'price': [78, 79, 0, 80, 81, 0, 82, 83, 0, 84]} Total qty =
0
34 : {'demand': 120, 'price': [0, 66, 67, 68, 69, 0, 70, 71, 0, 72]} Total qty =
0
35 : {'demand': 100, 'price': [82, 0, 83, 84, 85, 0, 86, 87, 88, 0]} Total qty =
0
36 : {'demand': 115, 'price': [0, 69, 71, 0, 72, 73, 74, 75, 0, 76]} Total qty =
0
37 : {'demand': 119, 'price': [77, 78, 79, 0, 80, 81, 82, 0, 83, 84]} Total qty =
0
38 : {'demand': 98, 'price': [0, 70, 71, 72, 0, 73, 74, 75, 0, 76]} Total qty = 0
39 : {'demand': 112, 'price': [81, 0, 82, 83, 84, 0, 85, 86, 0, 87]} Total qty =
0
40 : {'demand': 94, 'price': [0, 68, 69, 70, 0, 71, 72, 73, 74, 0]} Total qty = 0
41 : {'demand': 107, 'price': [83, 0, 84, 85, 0, 86, 87, 88, 89, 0]} Total qty =
0
42 : {'demand': 103, 'price': [0, 67, 69, 70, 71, 0, 72, 73, 0, 74]} Total qty =
0
43 : {'demand': 91, 'price': [79, 0, 80, 81, 82, 0, 83, 84, 85, 0]} Total qty = 0
44 : {'demand': 89, 'price': [0, 66, 67, 68, 0, 69, 70, 71, 0, 72]} Total qty = 0
45 : {'demand': 114, 'price': [76, 0, 77, 78, 0, 79, 80, 81, 82, 0]} Total qty =
0
46 : {'demand': 118, 'price': [0, 70, 72, 73, 0, 74, 75, 0, 77, 78]} Total qty =
0
47 : {'demand': 102, 'price': [81, 82, 0, 83, 84, 0, 85, 86, 87, 0]} Total qty =
0
48 : {'demand': 95, 'price': [0, 68, 69, 70, 71, 0, 72, 73, 0, 74]} Total qty = 0
49 : {'demand': 108, 'price': [78, 79, 0, 80, 81, 0, 82, 83, 84, 0]} Total qty =
0
50 : {'demand': 99, 'price': [0, 70, 71, 72, 0, 73, 74, 75, 0, 76]} Total qty = 0
51 : {'demand': 80, 'price': [70, 72, 0, 80, 68, 70, 0, 75, 77, 72]} Total qty =
80

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52 : {'demand': 85, 'price': [0, 65, 67, 0, 70, 72, 74, 0, 76, 70]} Total qty = 0
53 : {'demand': 90, 'price': [68, 0, 70, 72, 0, 75, 77, 79, 0, 74]} Total qty = 0
54 : {'demand': 95, 'price': [65, 67, 0, 70, 72, 74, 0, 76, 78, 0]} Total qty = 0
55 : {'demand': 100, 'price': [0, 66, 68, 70, 72, 0, 75, 77, 79, 81]} Total qty = 0
56 : {'demand': 80, 'price': [69, 0, 71, 73, 0, 76, 78, 0, 74, 77]} Total qty = 0
57 : {'demand': 85, 'price': [0, 67, 69, 0, 71, 73, 75, 0, 78, 80]} Total qty = 0
58 : {'demand': 90, 'price': [70, 0, 72, 74, 0, 76, 78, 80, 0, 75]} Total qty = 0
59 : {'demand': 95, 'price': [0, 68, 70, 72, 0, 74, 76, 0, 78, 80]} Total qty = 0
60 : {'demand': 100, 'price': [71, 0, 73, 0, 75, 77, 0, 79, 81, 0]} Total qty = 0
61 : {'demand': 80, 'price': [0, 66, 68, 70, 0, 72, 74, 0, 76, 78]} Total qty = 0
62 : {'demand': 85, 'price': [69, 0, 71, 73, 0, 75, 77, 79, 0, 72]} Total qty = 0
63 : {'demand': 90, 'price': [0, 67, 69, 71, 0, 73, 75, 0, 77, 79]} Total qty = 0
64 : {'demand': 95, 'price': [70, 0, 72, 0, 74, 76, 0, 78, 80, 0]} Total qty = 0
65 : {'demand': 100, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
66 : {'demand': 80, 'price': [71, 0, 73, 75, 0, 77, 79, 0, 72, 74]} Total qty = 0
67 : {'demand': 85, 'price': [0, 69, 71, 0, 73, 75, 77, 0, 79, 81]} Total qty = 0
68 : {'demand': 90, 'price': [70, 0, 72, 74, 0, 76, 78, 0, 75, 77]} Total qty = 6
69 : {'demand': 95, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
70 : {'demand': 100, 'price': [71, 0, 73, 0, 75, 77, 0, 79, 81, 0]} Total qty = 0
71 : {'demand': 80, 'price': [0, 66, 68, 70, 0, 72, 74, 0, 76, 78]} Total qty = 0
72 : {'demand': 85, 'price': [69, 0, 71, 73, 0, 75, 77, 79, 0, 72]} Total qty = 0
73 : {'demand': 90, 'price': [0, 67, 69, 71, 0, 73, 75, 0, 77, 79]} Total qty = 0
74 : {'demand': 95, 'price': [70, 0, 72, 0, 74, 76, 0, 78, 80, 0]} Total qty = 0
75 : {'demand': 100, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 95
76 : {'demand': 80, 'price': [71, 0, 73, 75, 0, 77, 79, 0, 72, 74]} Total qty = 0
77 : {'demand': 85, 'price': [0, 69, 71, 0, 73, 75, 77, 0, 79, 81]} Total qty = 0
78 : {'demand': 90, 'price': [70, 0, 72, 74, 0, 76, 78, 0, 75, 77]} Total qty = 0
79 : {'demand': 95, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
80 : {'demand': 100, 'price': [75, 0, 73, 0, 75, 77, 0, 79, 81, 0]} Total qty = 1
81 : {'demand': 80, 'price': [0, 66, 68, 70, 0, 72, 74, 0, 76, 78]} Total qty = 0
82 : {'demand': 85, 'price': [69, 0, 71, 73, 0, 75, 77, 79, 0, 72]} Total qty = 0
83 : {'demand': 90, 'price': [0, 67, 69, 71, 0, 73, 75, 0, 77, 79]} Total qty = 0
84 : {'demand': 95, 'price': [70, 0, 72, 0, 74, 76, 0, 78, 80, 0]} Total qty = 0
85 : {'demand': 100, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
86 : {'demand': 80, 'price': [71, 0, 73, 75, 0, 77, 79, 0, 72, 74]} Total qty = 8
87 : {'demand': 85, 'price': [0, 69, 71, 0, 73, 75, 77, 0, 79, 81]} Total qty = 0
88 : {'demand': 90, 'price': [70, 0, 72, 74, 0, 76, 78, 0, 75, 77]} Total qty = 7
89 : {'demand': 95, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
90 : {'demand': 100, 'price': [71, 0, 73, 0, 75, 77, 0, 79, 81, 0]} Total qty = 0
91 : {'demand': 80, 'price': [0, 66, 68, 70, 0, 72, 74, 0, 76, 78]} Total qty = 0
92 : {'demand': 85, 'price': [69, 0, 71, 73, 0, 75, 77, 79, 0, 72]} Total qty = 0
93 : {'demand': 90, 'price': [0, 67, 69, 71, 0, 73, 75, 0, 77, 79]} Total qty = 0
94 : {'demand': 95, 'price': [70, 0, 72, 0, 74, 76, 0, 78, 80, 0]} Total qty = 0
95 : {'demand': 100, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
96 : {'demand': 80, 'price': [71, 0, 73, 75, 0, 77, 79, 0, 72, 74]} Total qty = 7
97 : {'demand': 85, 'price': [0, 69, 71, 0, 73, 75, 77, 0, 79, 81]} Total qty = 0
98 : {'demand': 90, 'price': [70, 0, 72, 74, 0, 76, 78, 0, 75, 77]} Total qty = 0
99 : {'demand': 95, 'price': [0, 68, 70, 72, 0, 74, 76, 78, 0, 80]} Total qty = 0
100 : {'demand': 100, 'price': [71, 0, 73, 0, 75, 77, 0, 79, 81, 0]} Total qty = 0

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transporters

```

1 : {'cap': {1: 120, 2: 100, 3: 90, 4: 110, 5: 100, 6: 130, 7: 120, 8: 140, 9: 10
0, 10: 120}, 'cost': 4} Total qty = 0
2 : {'cap': {1: 80, 2: 70, 3: 60, 4: 90, 5: 80, 6: 100, 7: 85, 8: 95, 9: 70, 10:
80}, 'cost': 5} Total qty = 0
3 : {'cap': {1: 100, 2: 120, 3: 110, 4: 130, 5: 120, 6: 140, 7: 135, 8: 125, 9: 1
20, 10: 115}, 'cost': 6} Total qty = 0
4 : {'cap': {1: 90, 2: 100, 3: 80, 4: 110, 5: 120, 6: 130, 7: 120, 8: 140, 9: 11
0, 10: 120}, 'cost': 4} Total qty = 0
5 : {'cap': {1: 70, 2: 60, 3: 80, 4: 75, 5: 85, 6: 90, 7: 95, 8: 80, 9: 85, 10: 9
0}, 'cost': 3} Total qty = 0
6 : {'cap': {1: 100, 2: 120, 3: 130, 4: 110, 5: 125, 6: 140, 7: 135, 8: 120, 9: 1
25, 10: 130}, 'cost': 6} Total qty = 100
7 : {'cap': {1: 85, 2: 90, 3: 95, 4: 100, 5: 105, 6: 110, 7: 115, 8: 120, 9: 125,
10: 130}, 'cost': 5} Total qty = 0
8 : {'cap': {1: 60, 2: 65, 3: 70, 4: 75, 5: 80, 6: 85, 7: 90, 8: 95, 9: 100, 10:
105}, 'cost': 3} Total qty = 0
9 : {'cap': {1: 110, 2: 120, 3: 130, 4: 125, 5: 115, 6: 140, 7: 135, 8: 145, 9: 1
30, 10: 120}, 'cost': 6} Total qty = 0
10 : {'cap': {1: 100, 2: 105, 3: 110, 4: 115, 5: 120, 6: 125, 7: 130, 8: 135, 9:
140, 10: 145}, 'cost': 5} Total qty = 0
11 : {'cap': {1: 75, 2: 80, 3: 85, 4: 90, 5: 95, 6: 100, 7: 105, 8: 110, 9: 115,
10: 120}, 'cost': 4} Total qty = 0
12 : {'cap': {1: 90, 2: 100, 3: 110, 4: 120, 5: 130, 6: 140, 7: 120, 8: 110, 9: 1
00, 10: 90}, 'cost': 5} Total qty = 0
13 : {'cap': {1: 60, 2: 70, 3: 80, 4: 90, 5: 100, 6: 110, 7: 120, 8: 130, 9: 140,
10: 150}, 'cost': 4} Total qty = 0
14 : {'cap': {1: 130, 2: 125, 3: 120, 4: 115, 5: 110, 6: 105, 7: 100, 8: 95, 9: 9
0, 10: 85}, 'cost': 6} Total qty = 30
15 : {'cap': {1: 100, 2: 95, 3: 90, 4: 85, 5: 80, 6: 75, 7: 70, 8: 65, 9: 60, 10:
55}, 'cost': 3} Total qty = 0
16 : {'cap': {1: 90, 2: 100, 3: 110, 4: 120, 5: 130, 6: 140, 7: 135, 8: 125, 9: 1
15, 10: 105}, 'cost': 5} Total qty = 0
17 : {'cap': {1: 80, 2: 85, 3: 90, 4: 95, 5: 100, 6: 105, 7: 110, 8: 115, 9: 120,
10: 125}, 'cost': 4} Total qty = 0
18 : {'cap': {1: 100, 2: 110, 3: 120, 4: 130, 5: 140, 6: 150, 7: 135, 8: 125, 9:
115, 10: 105}, 'cost': 6} Total qty = 0
19 : {'cap': {1: 95, 2: 90, 3: 85, 4: 100, 5: 110, 6: 115, 7: 120, 8: 125, 9: 13
0, 10: 135}, 'cost': 4} Total qty = 0
20 : {'cap': {1: 70, 2: 80, 3: 90, 4: 100, 5: 110, 6: 120, 7: 130, 8: 140, 9: 15
0, 10: 160}, 'cost': 5} Total qty = 0
21 : {'cap': {1: 120, 2: 115, 3: 110, 4: 105, 5: 100, 6: 95, 7: 90, 8: 85, 9: 80,
10: 75}, 'cost': 3} Total qty = 0
22 : {'cap': {1: 100, 2: 105, 3: 110, 4: 115, 5: 120, 6: 125, 7: 130, 8: 135, 9:
140, 10: 145}, 'cost': 6} Total qty = 85
23 : {'cap': {1: 80, 2: 85, 3: 90, 4: 95, 5: 100, 6: 105, 7: 110, 8: 115, 9: 120,
10: 125}, 'cost': 4} Total qty = 85
24 : {'cap': {1: 60, 2: 70, 3: 80, 4: 90, 5: 100, 6: 110, 7: 120, 8: 130, 9: 140,
10: 150}, 'cost': 3} Total qty = 0
25 : {'cap': {1: 140, 2: 130, 3: 120, 4: 110, 5: 100, 6: 90, 7: 80, 8: 70, 9: 60,
10: 50}, 'cost': 5} Total qty = 80
26 : {'cap': {1: 100, 2: 110, 3: 120, 4: 130, 5: 140, 6: 150, 7: 135, 8: 125, 9:
115, 10: 105}, 'cost': 6} Total qty = 0
27 : {'cap': {1: 90, 2: 95, 3: 100, 4: 105, 5: 110, 6: 115, 7: 120, 8: 125, 9: 13
0, 10: 135}, 'cost': 4} Total qty = 0
28 : {'cap': {1: 80, 2: 85, 3: 90, 4: 95, 5: 100, 6: 105, 7: 110, 8: 115, 9: 120,
10: 125}, 'cost': 5} Total qty = 0
29 : {'cap': {1: 70, 2: 75, 3: 80, 4: 85, 5: 90, 6: 95, 7: 100, 8: 105, 9: 110, 1
0: 115}, 'cost': 3} Total qty = 0

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```

30 : {'cap': {1: 110, 2: 120, 3: 130, 4: 125, 5: 115, 6: 140, 7: 135, 8: 145, 9:
130, 10: 120}, 'cost': 6} Total qty = 0
31 : {'cap': {1: 100, 2: 80, 3: 90, 4: 0, 5: 110, 6: 95, 7: 100, 8: 0, 9: 105, 1
0: 90}, 'cost': 5} Total qty = 0
32 : {'cap': {1: 50, 2: 60, 3: 0, 4: 55, 5: 65, 6: 0, 7: 70, 8: 75, 9: 0, 10: 6
0}, 'cost': 4} Total qty = 0
33 : {'cap': {1: 80, 2: 0, 3: 85, 4: 90, 5: 95, 6: 0, 7: 100, 8: 105, 9: 110, 10:
0}, 'cost': 6} Total qty = 0
34 : {'cap': {1: 60, 2: 65, 3: 0, 4: 70, 5: 75, 6: 80, 7: 0, 8: 85, 9: 90, 10:
0}, 'cost': 5} Total qty = 0
35 : {'cap': {1: 90, 2: 95, 3: 100, 4: 0, 5: 105, 6: 110, 7: 0, 8: 115, 9: 120, 1
0: 0}, 'cost': 7} Total qty = 0
36 : {'cap': {1: 55, 2: 60, 3: 0, 4: 65, 5: 70, 6: 0, 7: 75, 8: 80, 9: 0, 10: 8
5}, 'cost': 4} Total qty = 0
37 : {'cap': {1: 70, 2: 0, 3: 75, 4: 80, 5: 85, 6: 0, 7: 90, 8: 95, 9: 0, 10: 10
0}, 'cost': 5} Total qty = 95
38 : {'cap': {1: 60, 2: 65, 3: 70, 4: 0, 5: 75, 6: 80, 7: 0, 8: 85, 9: 90, 10:
0}, 'cost': 6} Total qty = 0
39 : {'cap': {1: 50, 2: 0, 3: 55, 4: 60, 5: 65, 6: 0, 7: 70, 8: 75, 9: 0, 10: 8
0}, 'cost': 5} Total qty = 0
40 : {'cap': {1: 80, 2: 85, 3: 90, 4: 0, 5: 95, 6: 100, 7: 0, 8: 105, 9: 110, 10:
0}, 'cost': 7} Total qty = 0
41 : {'cap': {1: 55, 2: 60, 3: 0, 4: 65, 5: 70, 6: 0, 7: 75, 8: 80, 9: 0, 10: 8
5}, 'cost': 4} Total qty = 0
42 : {'cap': {1: 70, 2: 0, 3: 75, 4: 80, 5: 85, 6: 0, 7: 90, 8: 95, 9: 0, 10: 10
0}, 'cost': 5} Total qty = 75
43 : {'cap': {1: 60, 2: 65, 3: 70, 4: 0, 5: 75, 6: 80, 7: 0, 8: 85, 9: 90, 10:
0}, 'cost': 6} Total qty = 0
44 : {'cap': {1: 50, 2: 0, 3: 55, 4: 60, 5: 65, 6: 0, 7: 70, 8: 75, 9: 0, 10: 8
0}, 'cost': 5} Total qty = 0
45 : {'cap': {1: 80, 2: 85, 3: 90, 4: 0, 5: 95, 6: 100, 7: 0, 8: 105, 9: 110, 10:
0}, 'cost': 7} Total qty = 0
46 : {'cap': {1: 55, 2: 60, 3: 0, 4: 65, 5: 70, 6: 0, 7: 75, 8: 80, 9: 0, 10: 8
5}, 'cost': 4} Total qty = 75
47 : {'cap': {1: 70, 2: 0, 3: 75, 4: 80, 5: 85, 6: 0, 7: 90, 8: 95, 9: 0, 10: 10
0}, 'cost': 5} Total qty = 0
48 : {'cap': {1: 60, 2: 65, 3: 70, 4: 0, 5: 75, 6: 80, 7: 0, 8: 85, 9: 90, 10:
0}, 'cost': 6} Total qty = 0
49 : {'cap': {1: 50, 2: 0, 3: 55, 4: 60, 5: 65, 6: 0, 7: 70, 8: 75, 9: 0, 10: 8
0}, 'cost': 5} Total qty = 60
50 : {'cap': {1: 80, 2: 85, 3: 90, 4: 0, 5: 95, 6: 100, 7: 0, 8: 105, 9: 110, 10:
0}, 'cost': 7} Total qty = 80

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